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IoT Honeypot & Deception Framework (IHDF)

Project Submitted in Fulfillment of the CYB 496 Course requirements.

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Declaration

We Dunia Alshahrani, Layan Alshmmasi, Maysoon Alkhaldi, Sara Salman, Ohoud Alshehri, Buthaina Alhajlah, and Jana Aldawas, being members of final year project group number three, declare that this report contains only work completed by members of our group except for information obtained legitimately from literature, company or university sources. All information from these other sources has been duly referenced and acknowledged by the University Policy on Plagiarism.

Furthermore, we declare that in completing the project, the individual group members had the following responsibilities and contributed in the following proportions to the outcomes of the project:

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Abstract

The IoT currently connects billions of devices across the world to each other, smoothing several operations in homes, industries, and healthcare. The sudden development has increased the attack surface due to poor authentication and vulnerable devices. This project introduces the IHDF IoT Honeypot & Deception Framework, a lightweight security solution using deception for detection and analysis in IoT environments.

The framework deploys virtual IoT decoys that emulate real devices and communication protocols such as MQTT, REST, and RTSP. All the attacker interactions are captured and analyzed on a .NET backend, while a Flutter dashboard is used to display alerts and threat insights.

The results reveal that IHDF detects unauthorized access early, with reduced false positives and efficiency on limited resources. This work showcases the potential of deception-based defense in strengthening IoT security and enhancing real-time threat awareness.

Keywords: Internet of Things, Honeypot, Cybersecurity, Deception Framework, Smart Home

Abstract (in Arabic)

يرتبط إنترنت الأشياء اليوم ملائمة الأجهزة حول العالم، مما يساهم Np^0 تحسنت S^0 العمليات داخل المنازل والصناعات والقطاع المصرفي. لكن هذا التطور الهجوع أدى إلى توسيع كبر Np^0 1T سطح الهجوم تسبب ضعف آليات المصادقة وسهولة استغلال ثغرات الأجهزة. لتعرض هذا الم g^0 وع إطار العمل ودو إطار أم E^0 خفف الوزن لتعتمد على تقنيات الخداع للباشف عن الهجمات وتحللها Np^0 بثغرات إنترنت الأشياء، لتعتمد إطار العمل على تشغيل أجهزة افتراضية خادعة تحا p^0 الأجهزة الحقيقية و-بروتوكولات الاتصال مثل

MQTT وREST وRTSP.

، بينما تعرض التنبهات والنتائج.NET. يتم تسجيل جميع تفاعلات المهاجم S^0 وتحللها على نظام U^0 مطور للغة Flutter. من خلال لوحة تحكم مبنية باستخدام

تظهر النتائج أن النظام قادر على اكتشاف محاولات الوصول غير المصرح بها Np^0 مراحل مكررة، مع تقليل عدد الإنذارات الزائدة وتحسنت S^0 الأداء رغم محدودية الموارد. وهذا العمل فعال في استخدام تقنيات الخداع Np^0 تعلمها من إنترنت الأشياء وزيادة الوعي بالتهديدات لتشغل p^0 ... إنترنت الأشياء، الفخا 1^2 p^0 ، الأمن السيبراني p^0 ، هندسة الخداع، الم 1^0 q^0 ؛ البيانات المفتوحة

List of Abbreviations

MQTT	Message Queuing Telemetry Transport
API	Application Programming Interface
IoTPOT	Internet of Things Honeypot
USENIX WOOT'15	USENIX Workshop on Offensive Technologies, 2015
CPU architectures	Central Processing Unit Architectures (e.g., ARM, MIPS)
IoT	Internet of Thing
SSH	Secure Shell
Nmap	Network Mapper
Telnet	Telecommunication Network
ICS	Industrial Control Systems
TTPs	Tactics, Techniques, and Procedures
ML	Machine Learning
CoAP	Constrained Application Protocol
BLE	Bluetooth Low Energy
SCADA	Supervisory Control And Data Acquisition
MDP	Markov Decision Process
APT	Advanced Persistent Threat
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
HTTP	Hypertext Transfer Protocol

Table of Contents

Abstract.....	vi
Abstract (in Arabic).....	vii
List of Figures	xii
List of Tables	xiii
1 Introduction.....	2
1.1 Motivation	2
1.2 Problem Definition.....	3
Ideal Situation.....	3
Reality.....	3
Impact	3
State of the Art (solution)	3
Research Gap.....	3
1.3 State of the Art and Research Gap	4
1.4 Proposed Solution (optional)	5
1.5 Aims and Objectives	6
1.6 Project Scope	7
1.7 Report Structure.....	8
2 Background	11
2.1 Internet of Things (IoT).....	11
Industrial IoT (IIoT).....	12
Consumer IoT (CIoT)	12
2.2 IoT Security Challenges	13
2.3 Cyber Deception	14
Deception Technique	14
2.3.1 Role of Cyber Deception in Modern Security	16

2.4	Honeypot Technology	16
2.5	Machine Learning (ML)	17
3	Literature Review	21
3.1	Adaptive High-Interaction Honeypots in IoT Security.....	21
	Advantages of Adaptive High-Interaction Honeypots	22
	Limitations and Challenges	22
	Future Research directions	23
3.2	Deception Techniques for IoT Security.....	23
3.2.1	Comparative Analysis of Existing Studies	25
3.2.2	Strengths and Limitations of Current Deception Models	27
	Research Gap and Future Direction	27
3.3	Limitations of Machine Learning-Based IoT Honeypots and Future Research Directions.....	28
3.4	IoT High-Interaction Honeypot for Threat Analysis	30
	Advantages and Disadvantages	31
3.5	A Survey of Honeypots and Honeynets for Internet of Things, Industrial Internet of Things, and Cyber-Physical System	32
	Up with Changing Threat	32
	Challenge in Emulating IoT and IIoT Environments.....	32
	Hiding Honeypots from Attackers	32
	Big Data Storage and Analysis	33
	Real-World Testing Challenge.....	33
3.6	Adaptive Honeypot for Advanced Cybersecurity Threat Detection.....	33
3.6.1	Methodology	33
3.6.2	Findings.....	34
3.6.3	Limitations.....	34
3.6.4	Relation to the Project	34
3.7	Summary	35

4	System Analysis	40
4.1	System Perspective	40
4.2	Requirements Elicitation Techniques	41
4.2.1	Techniques Used and Findings.....	42
4.3	System Requirement	43
4.3.1	Use Case Diagram	44
4.4	Use Case Diagram Explanation	44
4.4.2	Non-Functional Requirements.....	49
4.5	System Interfaces.....	51
4.6	Summary	53
5	Proposed Solution	55
5.1	System Architecture	56
5.2	Methodology.....	Error! Bookmark not defined.
5.3	User Interface Design.....	60
5.4	Summary	61
6	Future Work	64
6.1	Project Timeline.....	66
	References	68

FIGURE 1 INDUSTRIAL IOT (IIoT) VS CONSUMER IOT (CIoT) [3]	11
FIGURE 2 CONCEPTUAL ARCHITECTURE OF CYBER DECEPTION FRAMEWORK FOR CRITICAL INFRASTRUCTURE (ADAPTED FROM [15])	15
FIGURE 3 INTEGRATION OF MACHINE LEARNING WITH HONEYPOT AND IDS IN IOT SECURITY FRAMEWORK.....	17
FIGURE 4 TAXONOMY OF INTRUSION DETECTION SYSTEMS IN IOT (SOURCE: A CRITICAL REVIEW OF IDS IN IOT, SPRINGEROPEN, [16])	18
FIGURE 5 EVALUATION FRAMEWORK FOR SMART CITY IOT HONEYPOT DEPLOYMENT (ADAPTED FROM [15])	26
FIGURE 6 IOT DECEPTION FRAMEWORK-USE CASE	44
FIGURE 7 SYSTEM INTERFACE OF THE IOT HONEYPOT & DECEPTION FRAMEWORK (IHDF) ADAPTED FROM [35].....	52
FIGURE 8 INTEGRATION OF HONEYPOTS WITH IDS IN AN IOT NETWORK ADAPTED FROM[15].....	55
FIGURE 9 OVERALL SYSTEM ARCHITECTURE OF THE PROPOSED IOT HONEYPOT FRAMEWORK, SHOWING THE INTEGRATED DECEPTION, ANALYSIS, AND COMMUNICATION LAYERS.....	59
FIGURE 10 ENTITY–RELATIONSHIP DIAGRAM (ERD) FOR THE IOT HONEYPOT & DECEPTION FRAMEWORK.....	60
FIGURE 11 IOT DECEPTION FRAMEWORK-HOME DASHBOARD	61
FIGURE 12 PROJECT TIMELINE (GANTT CHART)	67

List of Figures

List of Tables

TABLE 1 COMPARATIVE SUMMARY OF RECENT STUDIES ON DECEPTION TECHNIQUES FOR IoT SECURITY..... 26

TABLE 2 SUMMARY OF PREVIOUS STUDIES BASED ON TITLE, FRAMEWORK, KEY TECHNIQUES, FOCUS AREA AND MAIN FINDINGS
..... 37

TABLE 3 RELATIONSHIP SUMMARY 46

TABLE 4 USE CASE DESCRIPTION..... 46

TABLE 5 MAIN UML 2.0 SYMBOLS IN OUR DIAGRAM 47

Chapter 1

Introduction

1 Introduction

An IoT device is increasingly being seen as a target by attackers because of poor security practices such as use of default credentials, unsecured protocols, and lack of updates. To counter such a problem, one can resort to honeypot set-ups that emulate the real devices and collect data on attack attempts as well. This effectively allows researchers and practitioners to analyze the attacker's behavior and understand the emerging threats before the masses, and subsequently to build better defense strategies. This is precisely where our project intends to set up fake services for the early detection of threats and their analysis.

1.1 Motivation

The Internet of Things (IoT) is experiencing rapid growth, with projections indicating that there will be more than 29 billion connected devices by 2030 [16]. Although this growth facilitates the development of more intelligent homes, healthcare, and industries, it also significantly expands the cyberattack surface. Studies show that 57% of IoT devices are vulnerable to medium or high-severity attacks, and nearly 98% of IoT traffic is unencrypted, exposing sensitive data to attackers [17]. The Mirai botnet, a well-known incident, serves as a prime example of how the massive weaponization and hijacking of vulnerable IoT devices can occur [18].

Traditional defenses like firewalls and intrusion detection systems are often ineffective in IoT environments due to the devices' limited resources, lack of regular updates, and the challenge of detecting zero-day threats [19]. This motivates the need for innovative, lightweight, and proactive solutions. The IoT Honeypot & Deception Framework (IHDF) offers an effective approach by deploying decoy devices and services that attract attackers, enabling early detection, intelligence gathering, and reduced false positives. Such a framework is essential for protecting not only individual smart homes, but also larger critical infrastructures that rely on IoT ecosystems.

1.2 Problem Definition

Ideal Situation

The existing and upcoming IoT applications are highly promising to increase the level of comfort, efficiency, and automation for the users. To be able to implement such a world in an ever-growing fashion requires high security, privacy, authentication, and recovery from attacks.

Reality

IoT devices are constrained by low computational power, less memory and less energy which makes it difficult to implement complex security algorithms. Moreover, they are prone to several types of attacks such as DoS, malicious code injections, physical attacks and side channel attacks.

Impact

The security loopholes in IoT devices can be exploited to launch large scale attacks which may result in leakage of private information, financial loss and may also endanger human lives in certain critical applications like healthcare, smart transportation and industrial automation.

State of the Art (solution)

Several security architectures and mechanisms have been proposed for securing IoT which include lightweight encryption algorithms, authentication and access control mechanisms, intrusion detection systems, blockchain based solutions and machine learning based techniques.

Research Gap

Although several solutions have been proposed, most of them are either not scalable, too resource hungry for IoT devices, or not tested in real-world scenarios. Hence, there is still a need for comprehensive, lightweight and scalable security frameworks which can be practically implemented in large scale IoT deployments.

1.3 State of the Art and Research Gap

In IoT security, honeypots and deception techniques are widely used to study attacker behavior. Some common open-source honeypots like Conpot, Cowrie, and Dionaea are used to simulate services and collect attack logs. For IoT protocols, especially MQTT, there are some simple projects that provide decoy brokers to capture brute force logins or unusual topic access. In most cases, the data from these honeypots is stored as logs and then sent to dashboards such as ELK or Splunk for analysis. There are also some works that add fake admin portals or honeyfiles, but they are usually basic and not well connected.

Research Gap

Although these solutions exist, they still have some limitations:

Few solutions focus directly on MQTT decoys with clear detection rules.

- Existing tools are usually separated (one for decoys, another for logging, another for dashboard).
- Most honeypots provide low interaction and focus on a single service.
- Visualization is often limited to raw logs without showing attacker patterns clearly.
- Deployment can be difficult, especially in student or lab environments.

Position of Our Work

Our project, the IoT Honeypot & Deception Framework (IHDF), aims to solve these issues by:

- Providing IoT decoys such as MQTT broker, REST endpoints, and honeyfiles.
- Building a .NET backend with MySQL or Firebase to store and manage attack data.
- Adding simple detection rules like unauthorized login or suspicious topic access.
- Developing a React/Flutter dashboard for real-time monitoring of attacker interactions.

This framework will make it easier to deploy and analyze attacks on IoT systems, and at the same time give a more complete view compared to the existing solutions. That is why we decided to focus on MQTT and combine everything in one project, instead of using many separate tools.

1.4 Proposed Solution (optional)

The project proposes the design of the IoT Honeypot & Deception Framework to enhance detection capability for cyberattacks on IoT and critical infrastructure networks. The solution deploys honeypots and cyber deception assets to mislead attackers, gather intelligence, and offer early detection against them.

The framework will first deploy low-interaction honeypots, which provide fast detection for common attacks, like port scanning and brute force attempts, whereas high-interaction honeypots simulate real IoT devices and industrial services for deep analysis of attackers.

Second, the system will integrate decoy assets such as fake administrative portals, Honeyfiles-for instance, documents containing dummy sensitive data-and false credentials that are legitimate in appearance yet trigger alerts when accessed.

Thirdly, all the interactions of an attacker with decoys will get logged and subsequently sent to a central backend system. The backend will store event data that includes source IP addresses, frequency of requests, targeted protocols, and commands issued.

The logged information will then be used to gather threat intelligence and analysis, like identifying common attack vectors: unauthorized MQTT access, Telnet brute force, or API misuse. These analyses will help improve the defensive capabilities of IoT devices and could help patch vulnerabilities in real systems.

Finally, early warnings and response are facilitated under the framework. Since normal users never interact with deception assets, any access attempt signals malicious intent. On a dashboard, security administrators will be able to monitor attacker activity in real time, thus enabling fast incident response and mitigation. The proposed solution lays out an

extra layer of security in IoT environments against evolving cyber threats by combining honeypots, deception techniques, centralized monitoring, and analysis.

1.5 Aims and Objectives

Aim

The objective of the project is to design and deploy an IoT Honeypot & Deception Framework (IHDF) to protect Internet of Things (IoT) systems. The IHDF will leverage decoy devices and services that look authentic to capture, deter, and profile malicious traffic to mitigate the chances of system compromise and shed light on evolving threats.

Objectives

This project is designed to fulfill its aim through the following specific objectives:

- **Defining Threat Models:** Define and describe applicable threat trends in IoT environments, i.e., MQTT brute-force attacks and REST API exploitation.
- **High-Interaction Decoy Development:** Develop and deploy interactive decoy services—e.g., MQTT brokers, REST APIs, and honey files—to actively engage and lure attackers with decoy prospects.
- **Backend Infrastructure Deployment:** Deploy a secure backend infrastructure with .NET 8 Web API, and MySQL or Firebase to offer logging, data storage, and attack activity management with high efficiency.
- **Installation of real-time detection and analytics tools** involves the deployment of systems that can detect and analyze malicious activity or unauthorized access in IoT networks and offer advanced threat detection.
- **To create a dynamic analytics dashboard**, it is necessary to utilize tools like React or Flutter Web to create an interface that visually displays the metrics on the attack patterns, activity trend, and security incidents in such a way that they can be easily tracked and conclusions can be drawn.
- **Performance framework testing** involves the performance of rigorous testing through the employment of mock attack vectors in order to determine the

effectiveness of detection, the quality at which it gathers intelligence, and the solution stability when confronted with a wide range of attack vectors.

- Preparation of master documentation involves recording every project step, from strategy and outcomes of literature reviews to sequential deployment, details about prototypes, and deployment process in replicable and comprehensible form.

1.6 Project Scope

A Framework for Adaptive Honeypot Design and Evaluation, involves creating a comprehensive security framework focused on enhancing the protection of IoT environments using adaptive honeypots. Below is a breakdown of the project boundaries:

Application Type: The framework includes a software-based solution, particularly focusing on the deployment of honeypots within IoT ecosystems. This includes both physical and virtual honeypots to create deceptive environments for detecting and mitigating IoT-specific cyber threats.

Users: The primary users of this system will be cybersecurity specialists, researchers, and organizations involved in protecting critical infrastructure systems that rely on IoT devices.

Geographical Location: The project is not geographically restricted. Its application can be global, particularly relevant in areas with critical infrastructure reliant on IoT, such as smart cities, manufacturing plants, and utility systems.

Functionality: These honeypots dynamically adjust their behaviour based on observed attack patterns, using machine learning techniques to adapt.

Evaluation Framework: The system includes a methodology for assessing the effectiveness of honeypots, measuring metrics such as attack frequency, diversity, and intelligence gathering.

Integration with IDS: Honeypots are integrated with Intrusion Detection Systems (IDS) to enable proactive detection and analysis.

Dynamic Configuration: Leverages Software-Defined Networks (SDN) to create deceptive views specific to device groups.

The framework is designed specifically to protect critical infrastructure and is not intended for general consumer IoT devices unless they are part of a larger system within industrial or critical infrastructure settings.

This adaptive IoT framework represents a dynamic approach to addressing evolving cyber threats in interconnected systems, particularly those crucial for maintaining the operations of both public and private sectors.

1.7 Report Structure

Chapter 1: Almost all the information about the project is included in the background chapter. Such items as motivation, problem definition, aims and objectives, project scope and the proposed solution are presented. The background serves as a guide to grasping the project context and reasoning without going further.

Chapter 2: Background

The project is narrated in a very simple and clear way with the presentation of the most necessary concepts. The first technologies to be discussed are the Internet of Things (IoT), debunking of cyber technologies, honeypot technology, artificial intelligence, and the machine's monitoring systems (IDS). To make it clear, every subsection indicates the connection between the mentioned technologies and the security issues of IoT and counting the latter as the base of the proposed scheme. The background provides the necessary understanding of technology for the motivation and the design of the whole system.

Chapter 3: Literature Review

Earlier works and reports on IoT honeypots and deception frameworks have been reviewed in this chapter. An examination of adaptive high-interaction honeypots, deception techniques, machine learning applications, and related IoT security models has been performed. Moreover, the chapter points out the existing solutions' strengths and weaknesses, indicates such research gaps as limited scalability and lack of adaptive

learning, and finally, positions the proposed IHDF framework as a new approach that will successfully overcome these obstacles.

Chapter 4: System Analysis

The functionality aspect of the system is thoroughly discussed, and analytical views are given as well. The proposed honeypot-based defense is put side by side with traditional approaches such as IDS and firewalls, highlighting its proactive and intelligence-driven character. The chapter elaborates on requirement gathering techniques, specifies both functional and non-functional needs, and elucidates on system parts and application scenarios. In summary, it illustrates that the proposed system meets the determined research demands and thus, prepares the way for the framework's design and deployment.

Chapter 5: Proposed Solution

This chapter gives the complete proposed solution of IoT Honeypot & Deception Framework. It elaborates on the system architecture that consists of the network communication layer, protocol emulation, honeypot interaction modules, and the backend analytics and storage components in great detail. The chapter also represents the methodology employed, the ERD diagram, and the UI design. Overall, it provides clarity on the working of the whole framework and the roles played by the various components through deception technologies to enhance IoT security.

Chapter 6: Future Work and Project Timeline

This chapter covers not only the future work but also the project timeline which was prepared for the next phase. The future work part elaborates on the framework enhancements that are going to be made in the future like improving scalability, adding new protocols, making more realistic, supporting distributed honeynets, and integrating advanced learning mechanisms. Moreover, the chapter introduces the project timeline which narrates the expected schedule and task distribution for Graduation Project 2. This chapter indicates the long-term vision of the framework and arranges the execution of the upcoming development stages.

Chapter 2

Background

2 Background

In this section, we provide the fundamental background necessary to understand the proposed IoT deception-based defense framework. This includes an introduction to the concepts of Internet of Things (IoT), Cyber Deception, Honeypots, Machine Learning (ML), and Intrusion Detection Systems (IDS). These topics form the foundation of our work and help explain how deceptive mechanisms can enhance the security and monitoring of IoT environments.

2.1 Internet of Things (IoT)

The Internet of Things (IoT) integrates sensing, computation, and networked communication into physical objects so they can generate data, exchange it with services, and trigger actions. This creates a continuous loop (sense $\rightarrow$ transmit $\rightarrow$ analyze $\rightarrow$ act). Optimizing processes to increase safety and new service models in scale. We will discuss two categories IoT deployments namely Industrial IoT (IIoT) and Consumer IoT (CIoT) as seen in figure [2].

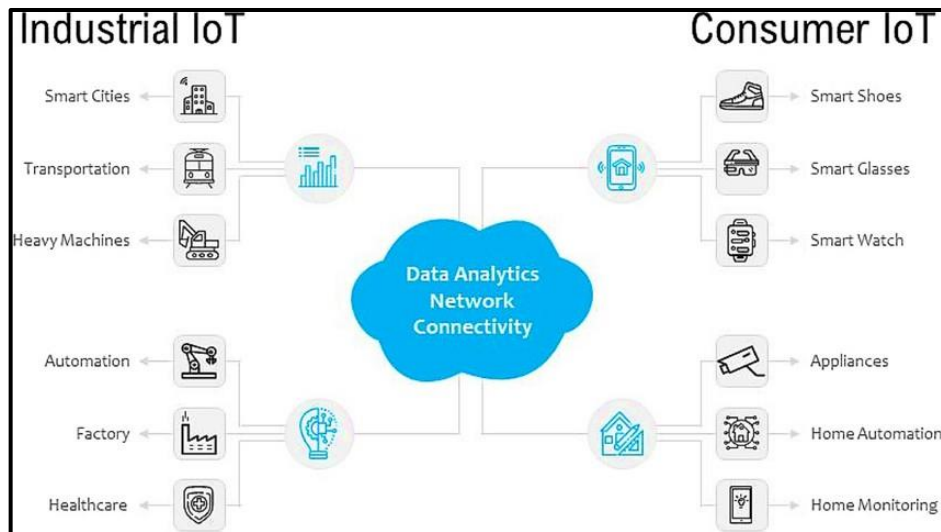


Figure 1 Industrial Iot (IIoT) VS Consumer Iot (CIoT) [3]

Industrial IoT (IIoT)

IIoT connects machines, production lines, and infrastructure to enable real-time visibility and control. Common applications are:

- Predictive maintenance of rotating assets such as motors, pumps, turbines.
- Compressor and HVAC plant condition monitoring to optimize energy consumption in factories and campuses.
- Digital twins of assembly cells and process lines for throughput tuning and anomaly detection.
- Asset tracking (UWB/RFID) in warehouses and yards to cut search time and shrinkage.
- Utilities and grids: smart meters, substation sensors, and demand-response controllers to balance load and reduce outages.
- Precision agriculture: soil-moisture nodes, fertigation valves, and weather stations coordinating irrigation and inputs.

These deployments depend on secure onboarding, reliable networking (Ethernet, Wi-Fi, private 4G/5G, LPWAN), edge gateways for protocol translation and cloud analytics for fleet-wide KPIs and maintenance scheduling [5].

Consumer IoT (CIoT)

CIoT enriches home, health, mobility, and retail daily lives. Common IoT daily scenarios are:

Wearables: Apple Watch and similar watch devices measure health indicators such as heart rate, ECG, blood oxygen, sleep. Also it streams activity data to enable alerts such as irregular rhythm notifications, personalized coaching, and integration with health records.

Smart homes: networked thermostats, lights, plugs, and blinds work together to manage comfort and energy. doorbells and cameras augment access control and security.

Home health: networked glucometers, blood-pressure cuffs, and spirometers allow remote monitoring and compliance.

Logistics: consumer tracking devices and telematics allow live location, geofencing, and item loss prevention for cars, bicycles, and packages.

Retail: Shelf sensors provide inventory visibility, dynamic pricing, and frictionless checkout.

Across IIoT and CIIoT, value is delivered by secure connectivity, over-the-air (OTA) update, and data-driven automation—turning distributed, resource limited devices into adaptive services that improve reliability, efficiency, and user experience [6].

2.2 IoT Security Challenges

Implementing honeypot technology within the IoT domain is extremely difficult. This is primarily due to the heterogeneity of IoT devices, i.e., vulnerabilities will vary immensely depending upon the brand name, model, or firmware version of the device. Therefore, attackers typically carry out a variety of pre-attack probes in order to get information about the destination device prior to launching an attack on it. For instance, Nmap [7], which is a widely used reconnaissance tool for scanning ports and identifying devices based on the output of probes, can also be used by attackers to determine if a device is a honeypot through the use of honeypot detection tools such as Honey Score [8] or Honeypot Hunter [9].

Traditional IoT honeypots [10] are limited in interaction since their response logic is fixed, and as a result, they cannot effectively mislead attackers during pre-attack checks. Honeypots have been used to defend against IoT devices attacks. Conventional IoT honeypots basically focus on emulating specific protocols such as telnet or ssh [10, 18]. The attacker may comprehend that a specific service is missing and may wonder if he is not speaking to a real IoT device. In addition, honeypot provides a limited level of interaction to attackers with some predefined. Playing logic (i.e., static answer to varying requests). These behavioral fingerprints might well have been picked up by the scanners [7, 11] used by the attackers to recognize the honeypots. Conventional IoT honeypots,

therefore, do not prevent the novel type of attackers. There is an urgent need to come up with adaptive high-interaction IoT honeypot that can interact with the attackers, bypass their pre-attack verifications, and mislead them into uploading their malicious codes.

2.3 Cyber Deception

Cyber deception is a proactive defense mechanism that aims to mislead attackers by creating false digital environments that appear legitimate. According to the paper *“Honeypot and Cyber Deception as a Tool for Detecting Cyber Attacks on Critical Infrastructure”*, deception allows security systems to anticipate and observe attacker behavior rather than only reacting to breaches after they occur. By embedding deceptive assets within the network—such as fake servers, administrative accounts, or decoy data—defenders can detect malicious activity at early stages and collect valuable intelligence about the attacker’s methods and goals. This concept represents a strategic shift from traditional reactive security toward intelligent, behavior-driven defense. Additionally, deception increases attackers’ cognitive load, forcing them to waste time validating false targets, which ultimately enhances system resilience [12].

Deception Technique

Numerous forms of cyber deception exist, each one decidedly crafted to manipulate an attacker's perception of the target environment. These techniques contain several constituents that, together, create a believable appearance of real assets within a network. The most basic form of deception is the honeypot: a system or service left intentionally open to attract attackers and monitor their activities. Honeypots can be anything from very simple emulations of common services like SSH, HTTP, or FTP up to high-interaction honeypots, which fully deploy an operating system. The analysis of the attacker's commands, payloads, and connection patterns will reveal new attack vectors and intrusion trends. Along with honeypots, honeyfiles are decoy documents placed in file systems and used by an organization to determine unauthorized access or sensitive data exfiltration attempts. Each honeyfile contains unique identifiers or is monitored by beaconing scripts that report back to the defenders if someone tries to open or copy the file. Thus, this

technique has been very useful for pinpointing insider threats and network lateral movement.

Another widespread deception mechanism involves decoy network services or virtual machines that mimic the properties of legitimate infrastructure components. Examples include phony web servers, IoT gateways, or administrative consoles used to entice attackers into interacting with fake targets. On interaction, metadata related to source IP, command sequences, and timing patterns are captured information that can later be used to train intrusion detection or machine learning models. Modern deception frameworks also combine both static and dynamic deception. While static deception depends on statically preset decoys, in dynamic deception, a system will adapt in real time-either by changing responses to the system or manufacturing new decoys-as an attacker explores the network. This allows for more realism toward the deceptions and prolongs the time of engagement by delaying attackers longer, therefore allowing more intelligence to be collected. According to the referenced paper, their deployment in critical infrastructure and IoT networks offers early warning capabilities often beyond that which is usually available with traditional intrusion detection systems. Deception serves not only as a detection layer but also as a learning environment in reinforcing the organization's understanding of adversary behaviour and vulnerable system weaknesses [12].

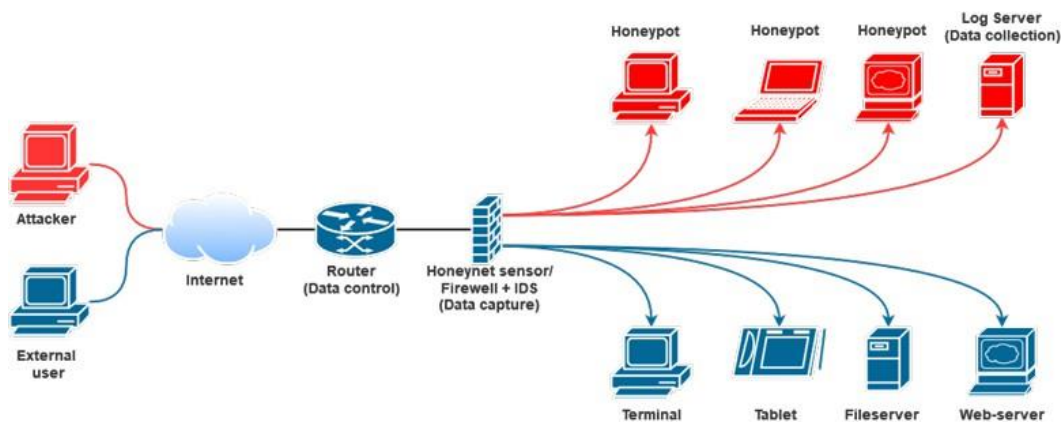


Figure 2 Conceptual Architecture of Cyber Deception Framework for Critical Infrastructure (adapted from [15]).

2.3.2 Role of Cyber Deception in Modern Security

Cyber deception fills in a very critical gap in current defense systems with detection in context and real-time attack analytics. Traditional intrusion detection systems depend on predefined rules or known signatures, whereas the mechanisms based on deception rely only on interaction-based evidence-any engagement with decoy resources means malicious intent is indicated. As discussed in the paper, this approach reduces false positives, increases incident response times, and feeds exquisite datasets to power future analysis or machine learning applications. This makes deception both a detection tool and a research tool to study an adversary's behavior in both IoT and industrial networks continuously [12].

2.4 Honeypot Technology

A honeypot is a decoy system perceived as a genuine device that works on attracting hackers and recording their behavior. So, it is really a trap that allows researchers to observe hacking activities with safety. Honeypots are particularly important in the realm of Internet of Things because so many smart devices are poorly secured, often using default passwords. By researchers setting up honeypots, real attempts of attacks can be perceived, Malware samples can be collected, and the ways in which hackers work toward attacking IoT devices can be studied.

In famous research, the IoTPOT project (USENIX WOOT'15) developed a honeypot emulating IoT devices through protocols such as Telnet. The system collected data from real attackers from all around the world, shining the spotlight on the rapid increase in IoT-targeted malware like Mirai. It also studied different CPU architectures (e.g., ARM and MIPS) employed by the IoT devices to investigate the means of malware propagation.

To sum up, honeypot technology enables IoT cybersecurity experts to detect new threats early, analyze hacker methods, and come up with improved methods for protecting the IoT environment. [13]

2.5 Machine Learning (ML)

Machine Learning (ML) plays a crucial role in enhancing modern cybersecurity mechanisms, particularly within IoT environments. It enables systems to automatically analyze large volumes of data, identify hidden attack patterns, and improve detection accuracy over time. In the context of IoT Honeypots and Deception Frameworks, ML can be applied to analyze attacker behavior, classify attack types, and predict potential future threats based on previous interactions.

by training models on the data collected from honeypot systems—such as login attempts, command sequences, and network traffic—ML algorithms can distinguish between normal and malicious activities more efficiently than traditional rule-based methods. This adaptive capability allows the framework to respond dynamically to evolving attack strategies. Moreover, integrating ML with intrusion detection and deception mechanisms strengthens the system’s ability to recognize zero-day attacks, reduce false positives, and enhance overall situational awareness.

Therefore, incorporating ML into the IoT Honeypot & Deception Framework supports intelligent decision-making and continuous learning, making the defense system more resilient, proactive, and capable of adapting to the ever-changing threat landscape of IoT environments [13].

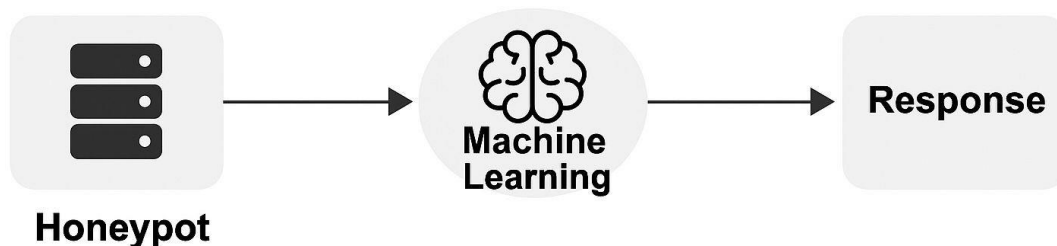


Figure 3 Integration of Machine Learning with Honeypot and IDS in IoT Security Framework.

2.6 Intrusion Detection Systems (IDS)

Intrusion Detection Systems (IDS) are invaluable security appliances that closely monitor system and network traffic activity in order to identify malicious or unauthorized activity. Think of an IDS as a sharp-eyed guard constantly scanning incoming and outgoing data for anomalies or known attack signatures. For IoT scenarios, IDS have particular problems: there are many lightweight devices using diverse protocols and may not have strong security parameters. Legacy IDS solutions, relying heavily on signature-based matching, are not effective against zero-day attacks and against novel attack patterns, not yet trapped in signature databases. To further enhance these constraints, researchers more and more incorporate anomaly-based approaches, machine learning, or combination approaches into IoT IDS frameworks. These techniques support detecting unusual behavior rather than employing pre-defined signatures, as depicted in Figure 4. Dwe

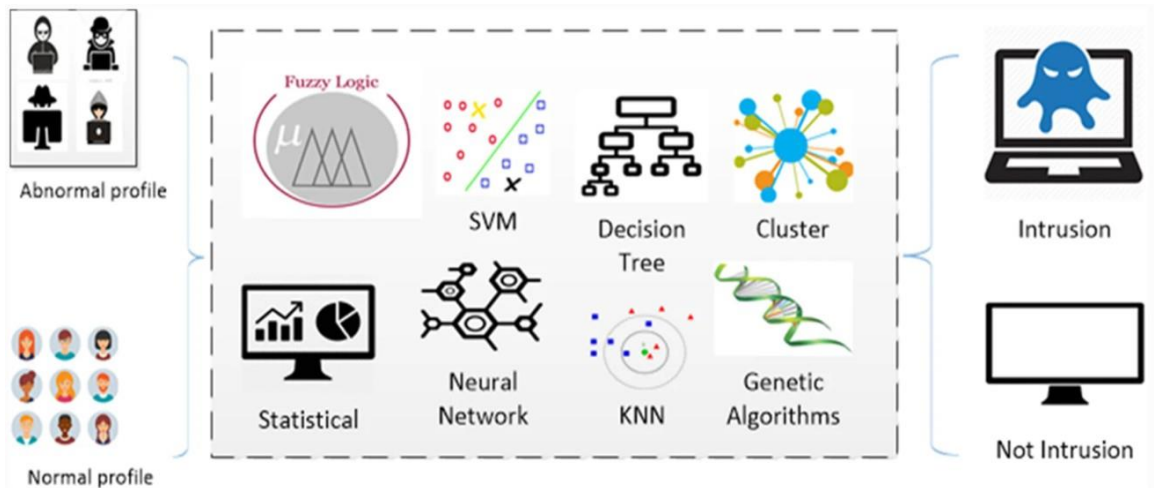


Figure 4 Taxonomy of Intrusion Detection Systems in IoT (Source: A Critical Review of IDS in IoT, SpringerOpen, [16])

Moreover, integration of deception and honeypots with IDS enhances detection capacity by trapping attacker action in protected systems. Information collected from honeypots is

used to raise awareness to the IDS about attacker activity so that the system learns from subsequent active attacks and increases future detection. Despite improvement, IoT-IDS systems are still plagued by false positives, encrypted traffic, limited computational capabilities, and heterogeneity of the data among devices. In short, IDS continues to be an integral part of the security of IoT infrastructures. Equipped with machine learning and deception-based methods, IDS can transform from reactive monitors to active defenders — identifying unknown attacks, learning from the attackers, and improving over time to match new threats.

Chapter 3

Literature Review

3 Literature Review

IoT honeypots and broader **deception frameworks** have become essential research tools for observing attacker behavior against connected devices, safely harvesting malware, and stress-testing defensive hypotheses in the wild. Recent surveys trace a shift from low-interaction decoys toward adaptive, high-interaction systems that better emulate real firmware, protocols, and device idiosyncrasies—reducing honeypot detectability and increasing intelligence yield. Deception now spans device-level lures, protocol impersonation (e.g., UPnP/MQTT/HTTP), edge gateways, and cloud backends, with machine-learning increasingly used to shape responses and automate engagement. This literature stresses two recurring design tensions: fidelity vs. cost (how “real” to make the device) and coverage vs. safety (how broadly to emulate without enabling harm). The following section will include a comprehensive review on related published work in 2023–2025 to synthesize these trends and position deception as a complement to detection/response pipelines for IoT botnet and exploitation research [8].

3.1 Adaptive High-Interaction Honeypots in IoT Security

Honeypots are now a highly effective threat analysis and prevention mechanism in the Internet of Things (IoT) scenario. Traditional low-interaction honeypots lack realistic attacker behaviors, causing sparse intelligence gathering. Existing research has shifted to adaptive and high-interaction honeypots that have the capability to emulate realistic IoT environments, provide continuous attacker interaction, and collect more nuanced threat data. Guan et al. introduced HoneyIoT, a honeypot system using dynamic honeypot that mimics device behavior through reinforcement learning application [9]. Representation of attacker–device interactions as an MDP enables HoneyIoT to adapt dynamically to send responses in real time, escaping reconnaissance checks and forcing attackers to deliver malware payloads. Experimental results demonstrated its ability to attract advanced threats and capture extensive attack trails, highlighting the viability and dependability of adaptive honeypot operations in IoT environments.

Furthermore, Javadpour et al. presented a comprehensive survey of cyber deception techniques that increase the validity and scalability of honeypots [10]. They concluded that machine learning, if integrated with deception-based approaches, has the ability to significantly improve response rates and reduce false positives over traditional intrusion detection systems. In another paper, Vetterl and Clayton built Honware, a virtual honeypot framework intended to collect IoT zero-day attacks by emulating consumer-premises equipment (CPE) and IoT devices [11]. Unlike HoneyIoT's emphasis on adaptability, Honware is centered on virtualization and scalability, making it possible for researchers to emulate different kinds of devices at low cost. In general, these papers show greater alignment of honeypot research towards intelligent, realistic, and scalable deceptions in IoT defense

Advantages of Adaptive High-Interaction Honeypots

Adaptive honeypots such as HoneyIoT provide several significant advantages. Their utilization of reinforcement learning helps them mimic real IoT activity more naturally, which makes them harder to identify as decoys [9]. It supports more prolonged attacker engagement and enables complete malware sample harvesting by attackers rather than mere shallow scanning attempts. Deception-based models discussed in [10] also boost detection credibility with higher behavioral realism and ongoing learning from the behavior of attackers. Also, virtualization platforms like Honware [11] demonstrate how scalable realistic honeypot behavior is achievable without costly physical hardware needs, thus allowing cost-saving research and broader deployment. Overall, these mechanisms effectively enhance threat visibility, data quality, and intelligence creation in IoT networks.

Limitations and Challenges

Although effective, adaptive and high-interaction honeypots are not without their limitations. HoneyIoT's learning depends on the timeliness and variety of training data, which may limit its flexibility to new device models or protocols [9]. Moreover, the operation of high-interaction honeypots introduces more complexity and risk since real

malware execution is facilitated that requires strong containment and isolation capabilities. Javadpour et al. noted that constant updating and retraining are essential to maintain deception systems, which could make them susceptible to detection as they become outdated [10]. Honware, while scalable, still uses significant virtualization resources to successfully emulate the behavior of various IoT devices [11]. These issues explain the realism, safety, and scalability compromise that should be maintained in adaptive honeypot design.

Future Research directions

Future research into adaptive honeypots focuses on improved scalability, automation, and support for multiple protocols. Guan et al. proposed expanding HoneyIoT to support other IoT protocols such as MQTT, CoAP, and UPnP, thereby increasing its use to industrial control and smart home systems [9]. Javadpour et al. recommended embracing transfer learning and adversarial training strategies in the creation of honeypots that autonomously adapt as the attacker strategy develops [10]. Further, leveraging the scalability of Honware [11], future deception systems can utilize cooperative honeynets—honeyproxied honeypots that simulate multiple devices interacting with one another to facilitate the analysis of lateral movement and coordinated attacks. Creation of standardized pipelines for converting collected honeypot telemetry to actionable detection rules will further bridge the gap between research prototypes and deployed security systems.

3.2 Deception Techniques for IoT Security

Recent findings indicate that adaptive honeypots dynamically adapt to observed attack patterns, which improves the accuracy of detection mechanisms [15]. Deception-based defense has gained attention as an advanced approach for protecting IoT environments from rapidly evolving cyber threats. Unlike intrusion detection systems based on pattern recognition, deception incorporates deliberate misinformation into the network to manipulate attacker behavior and acquire actionable threat intelligence. The idea of using honeypots or decoy servers and fake credentials that imitate legitimate IoT devices and services is a good example of such techniques. Careful monitoring of adversaries

interacting with such assets provides actionable information related to tools, tactics, and procedures. These can later be used to strengthen detection systems and improve the overall network resilience [14]. Morozov et al. [14] presented a comprehensive survey focused on cyber deception techniques that aim at improving the performance and efficiency of honeypots. Their work classifies deception mechanisms into static and dynamic categories—static deception includes preconfigured decoy systems while dynamic deception dynamically changes depending on the attacker’s strategy. This adaptability is particularly important in IoT ecosystems where device behaviors and traffic patterns constantly change. The study also pointed out the importance of lightweight deception layers that do not overload resource-constrained IoT devices, ensuring minimal disturbance with the maintenance of credible engagement for attackers.

Similarly, the authors of [15] proposed an adaptive honeypot framework specifically tailored for IoT networks. Their model develops an evaluation process capable of dynamically adjusting the honeypot's interaction level according to the sophistication of the detected attack. This approach provides the possibility to perform deeper analysis against sophisticated intrusion behaviors while reducing unnecessary waste of resources. Unlike general-purpose deception systems, Zhao's framework stresses contextual adaptation—that is, reacting to each attacker based on the nature of the IoT environment. This innovation improves the accuracy in intrusion detection and enhances situational awareness within distributed IoT infrastructures [15].

Collectively, these studies show that deception is not a passive monitoring tool anymore but an intelligence-driven and dynamic defense mechanism. With adaptive and layered deception, an IoT network can ensure the early detection of advanced threats, constantly learning from attacker behavior. This perspective aligns directly with the objective of the proposed project in designing a deception-based security framework capable of improving threat visibility and response efficiency in IoT ecosystems.

3.2.1 Comparative Analysis of Existing Studies

Both studies [14], [15] report on the use of deception as a defense strategy in IoT environments, although they approach the problem from slightly different angles. Morozov et al. [14] adopted a more general and theoretical viewpoint; the paper surveyed several methods of deception and groups them into static and dynamic classes of deception. Their work focuses on the enhancement of honeypot efficiency and realism through the implementation of adaptive behaviors in response to attacks.

However, their evaluation remains conceptual: it highlights what could be achieved, without testing the techniques in a live IoT environment. That makes their work valuable as a foundation to understand deception design but less practical when performance assessment is in order. On the other hand, Zhao et al. [15] developed and tested a hands-on framework, making their contribution more experimental and performance oriented. Their adaptive honeypot system was deployed on an IoT network and tested with real-world attack traffic.

The model dynamically adjusts its interaction level and logs detailed attack behaviors, showing stronger detection accuracy and a low false-alarm rate compared to static deception setups. While the work of Morozov provides a strong theoretical base, the system of Zhao proves how adaptive deception can work in practically realistic network conditions. Comparatively, Morozov's approach is more survey-based and analytical, while in contrast, Zhao's study is simulation-based and data-driven. Zhao's adaptive framework shows measurable improvements in terms of performance, achieving better detection accuracy and more efficient resource management for IoT devices. Regarding scope, Morozov's study applies to deception in general network security, while Zhao research directly targets IoT architecture, hence being more aligned with the direction of this project. Overall, both studies are complementary: [14] defines the conceptual structure of deception, while [15] translates that concept into a practical application in IoT. Taken together, they highlight how deception has evolved from a static idea of security to an adaptive, intelligent framework capable of facing current IoT threats. Table below presents a comparative summary of two selected studies in terms of their approach, performance, and application scope.

Table 1 Comparative summary of recent studies on deception techniques for IoT security.

Study	Method	Evaluation	Scope / Application	Key Findings
[14] Morozov et al., 2024	Survey & conceptual classification of cyber deception	Theoretical; no live IoT test	General network deception	Defines static vs. dynamic deception, highlights adaptability importance
[15] Zhao et al., 2024	Adaptive honeypot framework tested in IoT	Experimental; high detection accuracy, low false alarms	IoT-specific deception	Demonstrates real-time adaptation and efficient resource use

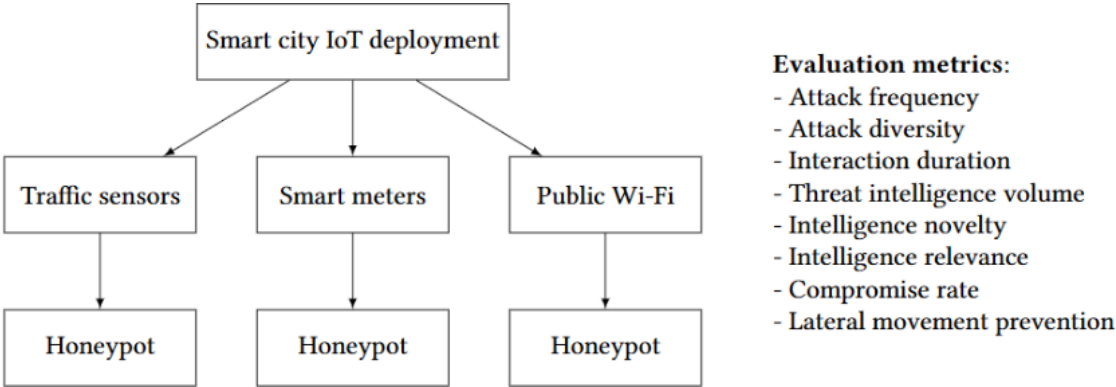


Figure 5 Evaluation framework for Smart City IoT honeypot deployment (adapted from [15]).

3.2.2 Strengths and Limitations of Current Deception Models

Studies [14], [15] have shown that deception-based mechanisms provide extensive benefits for IoT security. They allow for early detection of malicious activities, support detailed attack intelligence gathering, and provide threat isolation without exposing the operational systems. The adaptive honeypot frameworks improve detection accuracy through dynamic response mechanisms to attacker interactions [15]. These attributes make deception a viable method for IoT networks, which have limited computational resources. However, current models still have several limitations. Most the deception systems require extensive manual configuration and are hard to scale across heterogeneous IoT environments [14]. Another challenge is ensuring that decoy behavior keeps its authenticity; experienced attackers will quickly identify repetitive or unrealistic system responses. Furthermore, many of the frameworks have been validated only under artificial conditions in the laboratory, and not under operational conditions.

There is, therefore, a continuing need for adaptive, scalable, and lightweight deception frameworks which will be operational in an efficient way in large and diverse IoT infrastructures.

Research Gap and Future Direction

Although some recent works have advanced the use of deception in IoT security, there are still several limitations. Most frameworks are generalized and not fully optimized for IoT devices with limited power and memory [14], [15]. Besides, the collection of most prior works still often lacks large-scale validation and results from real-world deployment. Another gap in research is identified in the limited use of adaptive or intelligent techniques—such as machine learning—to complement the effectiveness of deception. Future studies should focus on developing lightweight, adaptive deception frameworks which can automatically adapt themselves to device behavior and network dynamics. These will be further evaluated under realistic IoT environments to measure scalability, performance, and long-term reliability.

3.3 Limitations of Machine Learning-Based IoT Honeypots and Future Research Directions

The paper titled *Optimizing Internet of Things Honeypots with Machine Learning* presents a general overview of the application of ML-based deception to IoT setups, but there are as yet some research gaps to be filled [16]. Most of the approaches discussed have been evaluated just under simulated conditions, which are not showing a realistic picture of the diverse and confined circumstances of real IoT gadgets [16]. In addition, there is very little knowledge about the computational expense and energy consumption of using ML models in honeypots, thus making it difficult for practical deployment [16]. Besides that, there is another limitation that comes from the non-existence of adaptive or self-learning deception strategies because the majority of the systems depend on static configurations for their operation [16]. Continuous learning is not applied to the systems that counterattack by evolving the behavior of the attacker [16]. Moreover, scalability and compatibility between various IoT platforms still remain unattended issues to be sufficiently treated [16]. Future investigations should aim at the development of compact and power-saving ML models that are capable of performing under the limitations set by embedded IoT devices [16]. It is also necessary to conduct real-world, long-term field experiments in order to assess the durability and the deception efficiency of such systems [16]. In addition, the use of reinforcement learning together with adversarial ML could equip the honeypots with the capability of dynamically adjusting [16].

3.3.1 System Architecture and Evaluation

The presented generic architecture of the ML-based honeypots consists of three notation components:

1. The Data Acquisition Layer gathers network interactions, system logs, and user activity traces all together.

2. Supervised and unsupervised ML algorithms come into play for labeling the attack behavior, finding anomalies, and suggesting the measures to be taken against the deception through the Learning Engine.

3. The Response & Deception Layer changes the honeypot's activity in real-time to either divert the attackers or activate the fake responses.

The experiment results point out that AI support has raised the detection correctness and threat prediction. On the other hand, this limitation due to the absence of small-footprint ML models applicable to IoT devices that are embedded greatly impacts its usage in edge or resource-constrained environments.

3.3.2 Advantages and Disadvantages of Machine Learning-Based IoT

Honeypots

Advantages: Through the adaptive pattern learning, the novelty or zero-day attacks are less likely to be detected at an early stage. The attack vectors can be automatically classified and the correlation of the IoT devices can be done easily. The deception can be made even with little human intervention as it is driven by the intelligence. **Disadvantages:** The complex training and inference steps cause a significant increase in computational overhead. Adaptability is limited as the models are unable to continuously learn or re-train based on live data.

The power and storage requirements limit the use of the technology for small IoT nodes. Theoretical and simulation-based experiments remain for most systems, which lowers their reliability in operations.

3.3.3 Research Gap and Future Work

The authors of the paper illustrate that the research in the area of ML for IoT security should be directed towards the creation of small-sized and energy-efficient ML models with the capability of deceiving the hackers. Besides, a fusion of reinforcement learning and adversarial ML could result in the generation of self-adaptive honeypots capable of discerning intruder actions and switching deception tactics in real-time. Moreover, it is necessary to conduct long-term trials in the field that would account for real-world attack

scenarios and thus, be able to measure performance and the sustainability of the model. The authors indicate that the future research directions should include interoperability between different platforms and the establishment of common performance metrics for the objective assessment of ML-based deception frameworks.

Eventually, this study brings to the forefront the issue of the IoT Honeypot & Deception Framework (IHDF) project, by pointing to the open question of the intelligent, adaptive, and resource-efficient honeypots that can silently and autonomously communicate in IoT ecosystems.

3.4 IoT High-Interaction Honeypot for Threat Analysis

Future IoT security research on honeypots emphasizes that high-interaction honeypots are important for analyzing even the most sophisticated cyberattacks. Unlike low-interaction honeypots, which just emulate a few services, high-interaction honeypots mimic the complete functionality of a device and full communication protocols. Such a design allows attackers to interact freely with the system, enabling the researchers to observe end-to-end attack chains, fetch real malware samples, and study post-exploitation behavior in isolated environments.

Elrawy et al. [17] implemented a high-interaction honeypot platform, intended for industrial IoT and smart home networks. The system uses real communication protocols-MQTT, CoAP, and HTTP-to emulate real IoT devices. It efficiently deceives attackers that exploit real device vulnerabilities and logs full sessions for inspection. Moreover, it allows the capture of live malware and forensic logging, providing contributions to research in advanced attack techniques and intrusion detection validation.

System Architecture and Evaluation

The system architecture embodies several key layers, such as protocol emulation modules, malware aggregation units, and a central analysis server. The protocol emulation modules-HTTP, CoAP, and MQTT-forward real-time attacker communications, while the central

analysis server does behavioral correlation in order to find out intrusion patterns. This honeypot has been used to capture a wide range of IoT attacks, which include unauthorized logins, botnet propagation, and malicious payload uploads in an experimental deployment, proving its effectiveness for gathering comprehensive threat data from real-world attack vectors.

Advantages and Disadvantages

Advantages: It captures comprehensive attack data in context, including complete attacker sessions, full command streams, malware samples, and post-exploitation activities. This is highly valued in APT analysis and for the creation of automated detection rules. **Disadvantages:** However, there are some real limitations due to the complexity and resource-intensive nature of the system: many full-protocol environments raise the risk of honeypot detection and possible containment breaches. The system does not include adaptive intelligence; it cannot change its responses dynamically according to the attacker's behavior, which reduces scalability and the effectiveness of the system in the long run.

Research Gap and Future Work

Although in the work of Elrawy et al. [17], high-interaction honeypots successfully demonstrate their potential in IoT threat research, several lines are identified for future improvement. For instance, there is a lack of adaptive learning mechanisms that provide the possibility of self-evolution against new attack strategies. Future development should be transferred to reinforcement learning with behavior-based deception in developing self-adaptive honeypots. Additional protocol coverage, especially with regards to Zigbee, BLE, and other proprietary IoT standards, would add more realism. Overall, the research provides a great foundation for intelligent, adaptive deception systems and has directly supported this project's objectives in establishing the IoT Honeypot & Deception

Framework (IHDF).

3.5 A Survey of Honeypots and Honeynets for Internet of Things, Industrial Internet of Things, and Cyber-Physical System

Up with Changing Threat

- **Sophisticated Attacks:** Honeypots in IoT and IIoT environments cannot keep pace with sophisticated attacks that change quickly. Attackers learn constantly how to identify and circumvent honeypots.
- **Targeted Attacks:** Targeted attacks that focus on particular system weaknesses can be tricky to identify with honeypots alone.

Challenge in Emulating IoT and IIoT Environments

- **High Heterogeneity:** IoT and IIoT devices are highly heterogeneous, from low-powered devices or devices with different architectures (e.g., low-energy processors or embedded systems). This poses a challenge to create full honeypots capable of emulating all device types.
- **Minimal Resources:** The majority of the devices in these environments have limited resources (e.g., memory, processing power), which makes it difficult to run sophisticated honeypots or even an effective honeynet.

Hiding Honeypots from Attackers

- **Detection of Honeypots:** Honeypots are simple for sophisticated attackers to detect unless made to appear realistic. This necessitates an ongoing need to develop honeypot designs that are interactive and more similar to real devices.
- **Need for Constant Updates:** Honeypots need constant updates to emulate the latest devices and systems in IoT and IIoT networks, which is an ongoing problem.

Big Data Storage and Analysis

- **Volume of Data:** Honeypots generate enormous amounts of data that need to be stored and analyzed. This requires high-end analysis and storage systems, which may not be suitable for resource-limited IoT and IIoT environments.
- **Effective Monitoring:** It may be complex to monitor the attacker's behavior from all this data and may require the application of techniques such as machine learning or AI to reduce data volume and improve analysis.

Real-World Testing Challenge

- **Real Environment Testing of Honeypots:** It is not easy to test the effectiveness of honeypots in real environments, since deploying honeypots on the actual system can negatively impact performance or security.
- **Honeynet Testing Limitations:** When configuring a honeynet to replicate a large network of machines, it's difficult to conduct real-world testing that guarantees the validity of the results in a secure environment.

3.6 Adaptive Honeypot for Advanced Cybersecurity Threat Detection

Muntaha et al. (2025) authored "Designing an Adaptive Honeypot for Advanced Cybersecurity Threat Detection," a system to improve cyber defense systems through adaptive deception. The study explored the efficacy of different honeypot interaction levels—low, medium, high, hybrid, and distributed—and how to combine them to produce realistic attack surfaces without affecting system efficiency. The main goal was to build a honeypot system that dynamically adapts to attackers' behavior and emerging threats.

3.6.1 Methodology

Researchers employed a hybrid approach with a combination of machine learning, reinforcement learning (DQN and DDQN), and game theory for real-time flexibility. The honeypot system was implemented using container-based technologies such as Docker and

Kubernetes to enhance scalability and isolation. Cloud deployment was also employed to evaluate the performance of adaptive honeypots in distributed large-scale environments.

3.6.2 Findings

Experimental results confirmed that adaptive honeypots significantly enhance detection effectiveness and attacker interaction compared to static honeypots. The system effectively protected against botnets, ransomware, and advanced persistent threats (APTs) by dynamically adjusting deception strategies based on attacker behavior. Interfacing reinforcement learning with containerization made it easy to modify the honeypot behavior with little downtime.

3.6.3 Limitations

In spite of its efficacy, the research recognized various challenges such as handling multiple dynamic containers' complexity and requiring ongoing model retraining to ensure adaptation precision. It also observed resource usage and scalability compromises upon scaling large-scale enterprise deployments.

3.6.4 Relation to the Project

This research directly adds to the current IoT Honeypot and Deception Framework project as both have a common theme of adaptive security solutions and innovative deception. The proposed method aligns with the project objective of developing an adaptive system that can enable attack data gathering, malicious activity investigation, and IoT network safeguard through ongoing learning and deception-based responses.

3.7 Summary

The papers reviewed showcase impressive developments in IoT deception and honeypot technology. The previous research has all sought to improve detection, data collection, and profiling of attackers by embracing static and dynamic deception strategies. The current trend is moving from traditional low-interaction honeypots towards adaptive and intelligent high-interaction systems which integrate artificial intelligence (AI) and reinforcement learning for simulating real device behaviors and entrapment of attackers

Based on studies perused (such as those by Morozov, Zhao, Elrawy, Muntaha, etc.), high-interaction honeypots are observed to trap richer information and more meaningful attacker behavior compared to low-interaction honeypots. Although issues persist regarding resource limitations, scalability, and encapsulation of security, literature also points out realism vs. safety, and coverage vs. cost trade-off.

In integrating the studies, the overall corpus of literature shows that adopting adaptive intelligence coupled with realistic device simulation helps considerably enhance the ability to harvest threat intelligence. Although theoretical summaries provided frameworks and taxonomies for IoT deception, experimental studies demonstrated the effectiveness of reinforcement learning and virtualization in improving detection accuracy and reducing false positives. Aside from these contributions, few studies provide large-scale or long-duration real-world experimentation.

Research Gaps Identified

- Lack of large-scale, long-term real-world testing: Most of the studies employ, simulation or laboratory experiments with no deployment.
- Need for lean AI models: Existing adaptive systems consume high-resources.
- incompatible with IoT's resource-starved platform.
- Limited protocol diversity: Very few studies for non-standard IoT protocols such as Zigbee, BLE, or bespoke systems.
- No continuous learning mechanisms: Most honeypots lack real-time adaptation or retraining for building attack plans.

- Lack of a consolidated evaluation metrics: There is no common platform used to compare the cost or effectiveness of deception systems
- Honeypot detectability risk: Attackers can identify and avoid honeypots because of known patterns.

Finally, Chapter 3 demonstrates that lot deception work is progressing towards intelligent, adaptive, and high-interaction honeypots with active interaction responses against attackers. However, there is still a gap in developing lightweight, scalable, and adaptive learning deceptions that would be deployable in secure and effective manners in real-world IoT settings. Future research would be focused on developing efficient AI-driven adaptive honeypots, broadening protocol coverage, and developing standardized performance measurement frameworks to determine effectiveness in real-world deployments.

Table 2 Summary of previous studies Based on Title, Framework, Key Techniques, Focus Area And Main Findings

No.	Paper Title & Year	Approach / Framework	Key Techniques	Focus Area	Main Findings
1	HoneyIoT: Adaptive High-Interaction Honeypot for IoT Devices Through Reinforcement Learning (WiSec 2023)	Adaptive high-interaction honeypot using reinforcement learning	Markov Decision Process (MDP), differential response mutation	Realistic IoT emulation and adaptive deception	Developed an adaptive honeypot that learns attacker behavior from traces, bypassing honeypot detection tools like Nmap and HoneyScore.
2	IoTPOT: Analysing the Rise of IoT Compromises (USENIX WOOT 2015)	Telnet-based honeypot and sandbox (IoTPOT + IoTBOX)	Multi-CPU architecture emulation (ARM, MIPS, PPC), malware sandboxing	Large-scale collection of IoT malware	Identified 4 major DDoS malware families and analyzed propagation via Telnet; demonstrated scalability of multi-arch honeypots.
3	Honware: A Virtual Honeypot Framework for Capturing CPE and IoT Zero Days (IEEE 2019)	Virtual high-interaction honeypot for firmware emulation	Firmware image emulation, Linux kernel customization	CPE & IoT zero-day detection	Captured new DNS-changing zero-day attacks; improved realism using firmware-based emulation without physical hardware.
4	Optimizing IoT Honeypots with Machine Learning: A Systematic Review (MDPI 2025)	Systematic literature review	Supervised, unsupervised, and reinforcement learning	ML-driven honeypots, detection optimization	ML enhances accuracy (up to 0.96) and adaptability; hybrid honeypots with ML reduce false positives and improve resource use.
5	IEEE Definition of IoT (IEEE 2015)	Foundational architecture definition	Standardization overview (IEEE, IETF, ITU)	IoT conceptual model	Defines IoT as a distributed network of smart objects with communication, management, and self-optimization capabilities; baseline for IoT frameworks.
6	Honeypot and Cyber Deception as a Tool for Detecting Cyber Attacks on Critical Infrastructure	Deception-based intrusion detection	Cyber deception layers, critical-infrastructure modeling	Threat detection & response	Highlights honeypots as an essential layer for detecting advanced persistent threats (APT) in industrial systems.

	(Conference 2024)				
7	A First Step Towards Understanding Real-world Attacks on IoT Devices (arXiv 2020)	Multi-stage honeypot ecosystem	Low to high-interaction design, cloud deployment	Real-world IoT attack capture	Demonstrated real-time human interaction with honeypots; introduced “HoneyCamera” for realistic IoT camera deception.
8	The Sweet Taste of IoT Deception: An Adaptive Honeypot Framework for Design and Evaluation (Journal of Edge Computing 2024)	Adaptive deception framework	Dynamic honeypot architecture, evaluation metrics	Adaptive honeypots for critical infrastructure	Proposed a two-component adaptive system that dynamically adjusts behavior and measures deception effectiveness quantitatively.

Chapter 4

System Analysis

4 System Analysis

The suggested Honeypot is aimed at attracting and trapping malicious attackers through the simulation of real network environments. All activities, including IP addresses, attack patterns, and tools used, are tracked upon the intruder's interaction with the system. Thus, cybersecurity analysts will be able to analyze attacker behavior based on the data collected to enhance general network security.

The provided analysis presents comprehensive details on the functionality and architecture of the system, thus proving that it solves the identified research problem. Further, it covers the existing research gap by presenting an intelligent, adaptive Honeypot system that can detect cyber-attacks in real time, thus paving the way for understanding attacker behavior without actually exposing systems to such risks.

4.1 System Perspective

Compared to traditional IDS or firewalls, the Honeypot reveals a much more proactive and intelligence-driven defense capability. While firewalls tend to block suspicious traffic based on predefined rules, and IDS systems monitor network activities to raise alerts when abnormalities are detected, a Honeypot directly confronts the attacker. It is an interactive trap that tricks adversaries into believing they have found a legitimate target. In this way, the system captures a complete and realistic image of the attacker's methods, techniques, motivations, and behavior—a dataset unavailable for traditional security solutions because they are designed to avoid rather than encourage contact.

While most IDS technologies struggle with false positives and often miss sophisticated attacks that blend into normal traffic, a Honeypot is devoid of such limitations since any activity against it is considered inherently suspicious, as no legitimate user would have a reason to access it. This leads to extremely high-quality threat intelligence that allows defenders to understand emerging attack vectors and prepare proactive countermeasures rather than simply rely on reactive alerting.

Moreover, the proposed system exhibits advanced automation mechanisms and multi-layer data correlation as opposed to previous generations of Honeypot systems, which only logged basic connection attempts or some simple steps of exploitation. The system reconstructs whole attack chains from analyzed connections, behavioral sequences, command patterns, and file interactions instead of gathering raw logs in isolation. Thus, it is able to identify complex intrusions that come in multi-stages, such as privilege escalation, lateral movement attempts, persistence strategies, and command-and-control communications.

Thus, with such an improved capability for correlations, the system interprets the attacks instead of only observing them and gives cybersecurity analysts the ability to trace the intentions of the attacker, identifying the sophistication level of the threat actor or mapping his behavior to known frameworks like MITRE ATT&CK. In fact, this enhances Honeypot from a passive logging device to a strong analytical tool, capable of incident response, vulnerability assessment, and strategic cyber defense planning.

4.2 Requirements Elicitation Techniques

Requirements elicitation represents an important step in system analysis, oriented to the identification and gathering of information on what a system is supposed to achieve and under which conditions it is supposed to operate.

The main objective of this process is to make sure that the final system design will meet user expectations, technical constraints, and the overall goals of the project. In cybersecurity-related projects, especially those involving IoT environments, requirements elicitation plays a vital role in understanding both functional needs related to attack monitoring and adaptive response, as well as non-functional needs concerning scalability, performance, and reliability. Requirements elicitation provided the grounds for defining how the proposed IoT deception-based framework should behave, what components it must include, and which performance standards it should meet to enhance the protection of the network through deception mechanisms.

4.2.1 Techniques Used and Findings

The two primary techniques employed toward accurately determining the requirements of the proposed IoT deception-based security framework are the literature review and comparative analysis of the existing systems.

Literature review remains the main foundation for the identification of the system's essential requirements. From previous works on IoT deceptions and honeypot-based defence mechanisms, the research team could extract the most relevant features and security needs which included adaptive honeypot deployment, continuous monitoring of the attacker's behaviour, automated data logging, and scalable response mechanisms. Reviewing recent academic work also helped this team understand the challenges faced in previous frameworks which include resource limitations among IoT devices and the maintenance of realistic deception interactions. Secondly, comparative analyses were conducted among various existing IoT deception architectures and honeypot frameworks. Such a comparison was needed to identify strengths and weaknesses regarding adaptability, integration with IoT protocols such as MQTT and CoAP, and general efficiency. In this respect, the analyses showed that most existing systems cannot maintain scalability and adaptability when deployed in heterogeneous IoT environments.

Further, limited communication compatibility and high resource consumption were among the common issues that reduced effective deception-based defence. These findings directly influenced the formulation of the proposed system's requirements. For instance, the new framework emphasizes lightweight architecture, adaptive deception behaviour, and improvement in protocol integration to ensure practical deployment in IoT ecosystems. The insights obtained from both literature and comparative analysis thus provided a concrete basis for defining the functional and non-functional requirements discussed in the next section.

The outcome of the elicitation process gave a clear direction for the system's development. These findings have helped in prioritizing key aspects of the proposed IoT deception framework in terms of adaptability, efficiency, and integration with communication

protocols used in the IoT environment. These insights have shaped both the functional and non-functional requirements defining how the system should operate and perform.

The following section elaborates on these various requirements, thereby laying the foundation for the design and implementation of the system.

4.3 System Requirement

This section presents the system requirements of the proposed IoT Honeypot & Deception Framework. It outlines the necessary functions the system needs to execute, along with the quality attributes guiding its behavior. The requirements are separated into two categories: functional requirements, which describe the main operations of the system, and non-functional requirements, which refer to performance, scalability, and security. These requirements provide the basis for the design and implementation of the system.

4.3.1 Use Case Diagram

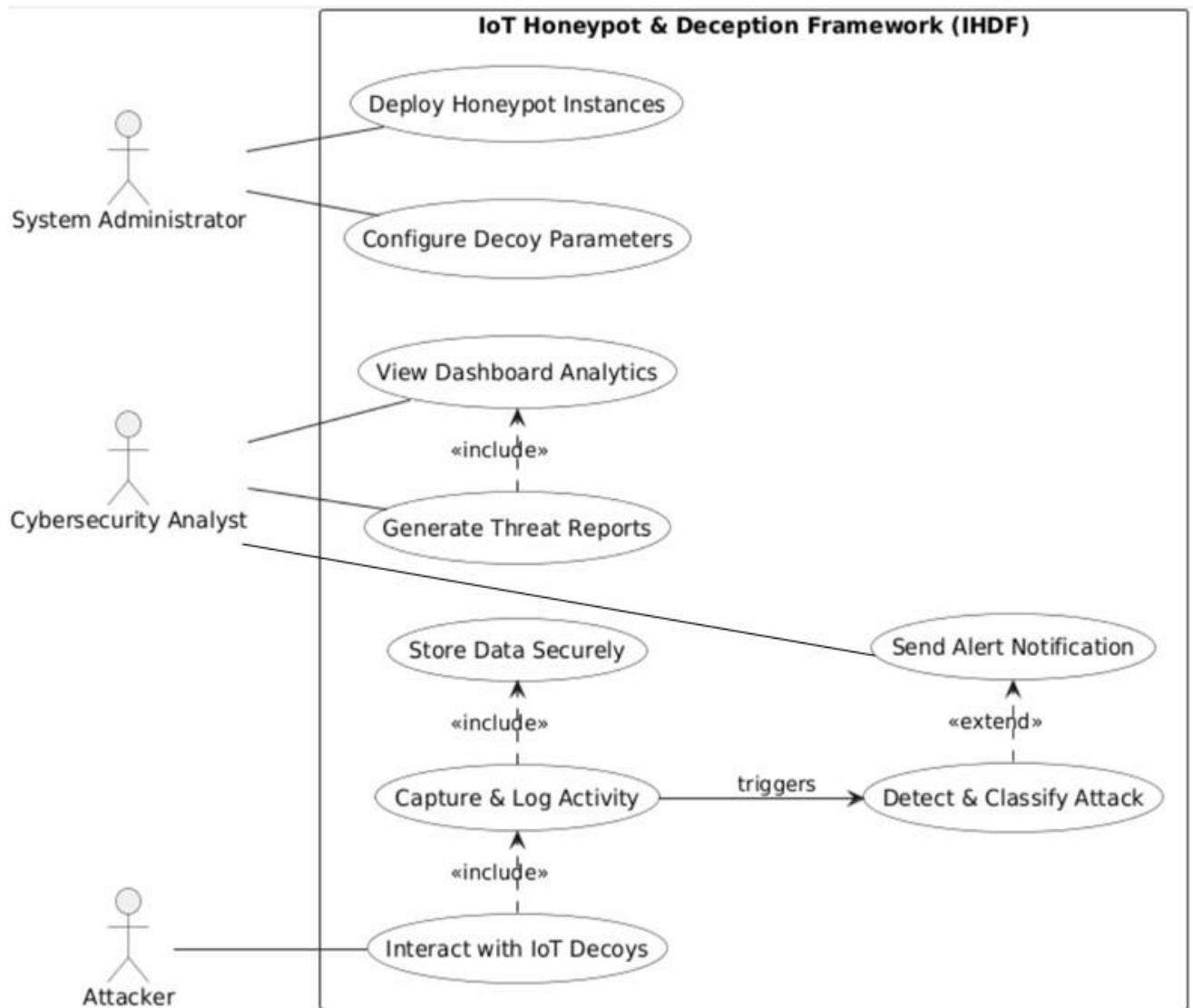


Figure 6 IoT Deception Framework-Use Case

4.4 Use Case Diagram Explanation

Figure X's Use Case Diagram shows how the IoT Honeypot & Deception Framework (IHDF) works with its main users: the System Administrator, the Cybersecurity Analyst, and the Attacker. It explains how each player uses the system to do important things like setting up honeypots, looking for threats, and finding bad behavior. The graphic employs

UML 2.0 symbols like "include," "extend," and "trigger" to show how use cases are related to each other.

System Administrator

The System Administrator manages the honeypot environment. Their main functions are:

- **Deploy Honeypot Instances:** Introduces fresh IoT network honeypot devices.
- **Configure Decoy Parameters:** To make decoys realistic, these parameters specify communication settings, ports, and device behaviors.

By taking these steps, the deception framework is effectively maintained and customized.

Cybersecurity Analyst

The System Administrator manages the honeypot environment. Their main functions are:

- **Generate Threat Reports:** Produces summaries of attacks and system activity.
- **View Dashboard Analytics:** Displays real-time statistics and performance metrics.

The 'View Dashboard Analytics' use case is included in 'Generate Threat Reports' because reports contain visual analytics.

Attacker

The Attacker represents a malicious user who interacts with fake IoT devices. Their actions activate the system's detection processes:

- **Interact with IoT Decoys:** Attempts to access or exploit the decoy devices.
- **Capture & Log Activity:** Records all attacker actions.
- **Store Data Securely:** Saves logs and data for later analysis.
- **Detect & Classify Attack:** Analyzes and identifies the type of attack.

- **Send Alert Notification:** Extends the detection process and occurs only when a serious threat is identified.

Relationship Summary:

Table 3 Relationship Summary

Relationship	Description	Example
Include	A required process that happens as part of another.	Capture & Log Activity includes Store Data Securely
Extend	An optional process that occurs under certain conditions.	Send Alert Notification extends Detect & Classify Attack
Trigger	One process directly causes another.	Capture & Log Activity triggers Detect & Classify Attack

Use Case Description:

Table 4 Use Case Description

Actor	Use Case	Description
System Administrator	Deploy Honeypot Instances	Starts new honeypot devices in the network.
System Administrator	Configure Decoy Parameters	Adjusts settings and communication protocols for decoys.
Cybersecurity Analyst	Generate Threat Reports	Creates detailed summaries of detected attacks.
Cybersecurity Analyst	View Dashboard Analytics	Displays charts and system statistics.

Attacker	Interact with IoT Decoys	Tries to access fake IoT devices.
System (IHDF)	Capture & Log Activity	Records attacker actions and behaviors.
System (IHDF)	Store Data Securely	Saves collected data safely for analysis.
System (IHDF)	Detect & Classify Attack	Identifies and categorizes the attack type.
System (IHDF)	Send Alert Notification	Sends alerts when a high-severity threat is detected.

Main UML 2.0 Symbols in our Diagram:

Symbol	Meaning	Used For
Actor	A person or external system that uses our system.	System Administrator, Cybersecurity Analyst, Attacker
Use Case	A function or action that the system performs.	Deploy Honeypot, Detect Attack, Generate Report
System Boundary	The box that shows what belongs inside the IHDF system.	IoT Honeypot & Deception Framework
«include»	A process that is always part of another one.	Capture & Log Activity includes Store Data Securely
«extend»	A process that happens only in special cases.	Send Alert Notification extends Detect & Classify Attack

Table 5 Main UML 2.0 Symbols in our Diagram

4.4.1 Functional Requirements

These are the functional requirements that IHDF is expected to meet to have effective deception, adaptive intelligence, and comprehensive threat data collection within IoT ecosystems. They are derived from the core principles of adaptive honeypot architecture and deception-based cybersecurity systems.

4.4.1.1 Context-Aware and Adaptive Architecture

The system shall implement a context-aware honeypot architecture that will be able to dynamically change its behavior and adapt in accordance with the detected attack patterns and adversary interaction. It will make sure that the deception layer changes in real time, thereby increasing the attractiveness of the environment for the attackers and reducing fingerprinting or early detection.

4.4.1.2 Advanced Protocol and Device Emulation

The IHDF shall provide advanced emulation capabilities for the most common IoT-communication protocols such as Telnet, SSH, HTTP, and MQTT. It shall simulate realistic responses of IoT devices, configurations, and vulnerabilities—for example, default credentials or open ports—with the purpose of attracting attackers. A medium-interaction level should be allowed—exposing a limited execution of commands and data exchange—to make it so the system gathers intelligence without exposing the infrastructure in the back

4.4.1.3 Centralized Management and Adaptive Reconfiguration

Central to this would be the management controller that would oversee all honeypot

instances in a coordinated deployment and adaptive reconfiguration. This includes auto-addition, removal, or modification of honeypots based on attack trends, thus optimizing coverage and maintaining deception credibility across diverse IoT environments.

4.4.1.4 Completa Acquisition de Datos y Logging Seguro

The IHDF must provide for the continuous, secure collection of all attacker interaction data coupled with network traffic to support in-depth analysis and intelligence generation. This includes complete packet-level traffic capture in PCAP format for forensic analysis, while at the same time recording every executed command with timestamps and all session metadata in order to keep a detailed operational trace of what the attackers do. It should also store malicious uploads like binaries or scripts in a sandbox-isolated environment to prevent any system compromise. These combined measures ensure traceability, forensic readiness, and the integrity of all intelligence gathered by means of this framework.

4.4.1.5 Automated Threat Intelligence and Reporting

The solution shall automatically analyze captured data and generate actionable threat intelligence by correlating honeypot events with the behavioral indicators to extract IoCs, like source IPs, payload signatures, and attacker credentials. It should also create real-time alerts and present analysts with in-depth reports of the summarized attack vectors, frequency, and behavior trends through a unified dashboard.

4.4.2 Non-Functional Requirements

The IoT Honeypot & Deception Framework must meet the non-functional requirements in order to guarantee realism, safety, scalability, and efficient operation. These characteristics are crucial for building a trustworthy and convincing honeypot environment because they emphasize how the system should behave rather than what it does.

Adaptability and Scalability:

Honeypots must be able to effectively mimic a variety of device types and protocols as the diversity and complexity of IoT devices continue to increase [26]. In order to adapt to changing IoT ecosystems, this calls for modular, configurable architectures.

Realism and Fidelity:

Honeypots must closely resemble the behavior and features of actual IoT devices in order to successfully engage and trick attackers [1]. This covers both network protocols and device-specific features like actuation capabilities and sensor data generation.

Containment and Safety:

To prevent honeypot compromise from impacting real systems, strict network isolation and device hardening are essential, especially for physical honeypots in ICS with potential physical consequences.

Adaptability:

To match changing IoT ecosystems, honeypots should facilitate simple swapping of emulated devices/protocols (container and SDN technologies can aid such flexibility) 5. Specialized Interaction

Expert Communication :

To engage attackers and learn domain-specific TTPs, honeypots should implement both generic protocols and IoT/ICS-specific functionality, such as device pairing or control system alarming.

Generation of Intelligence :

We use both custom components and open-source tools to implement our adaptive IoT honeypot framework.

The framework's adaptive decision-making components are implemented using the Python machine learning library Scikit-learn [27]. In order to forecast attacker intent and dynamically modify the honeypot configuration, we intend to train models on past attack data [15].

Sustainability :

Maintenance: Easy, requires little configuration; Moderate, requires script updates; Difficult, requires ongoing patching and monitoring.

High-interaction honeypots [4] give attackers access to real operating systems and collect detailed data, but if not properly contained, they run the risk of being used to attack other systems.

- **Metrics (Verification/Measurability)**

These metrics were selected to fully address the main goals of honeypot deployment, such as intelligence gathering (volume, novelty), containment capabilities (compromise rate, lateral movement prevention), and attacker engagement (attack frequency, diversity, duration). Honeypot performance can be quantitatively measured and benchmarked using the given formulas. Evaluating the effectiveness of honeypot and honeynet deployments in Internet of Things environments is crucial. We recommend the following framework and metrics for assessing IoT honeypot performance.

In conclusion, these non-functional requirements ensure the safe operation, effective adaptation, and realistic, measurable performance of the IoT Honeypot & Deception Framework. Together, they increase the system's resilience and strengthen its detection and mitigation of IoT-based threats.

4.5 System Interfaces

The proposed IoT Honeypot & Deception Framework (IHDF) system interfaces that provide communication, control, and data management among users, honeypots, and analytical components. These interfaces not only help in the easy integration but also in the security and usability of the whole system.

User Interface (UI): The User Interface is a web-based dashboard that is easy to use and shows the latest status of honeypots. Besides, it allows the administrators and security analysts to see the main features like visual analytics, deployment options, and automatic reporting tools. The design is mainly aimed at delivering an easy-to-use interface along

with secure access which is made possible by the use of authentication controls and encrypted sessions for authorized users.

System-to-System Interface: A secure connection can be made between the honeypot nodes, the central management controller, and the IoT devices using protocols such as MQTT, CoAP, and HTTP. Besides, it conducts real-time synchronization of attack data, and dynamic reconfiguration of the honeypots occurs according to the behavior of the attacker. Encrypted APIs and message queues are the only ways to communicate within the system, which indicates that the system's secrecy and data quality have always been preserved.

Figure 5: System Interfaces of the IoT Honeypot & Deception Framework (IHDF) adapted from [35]

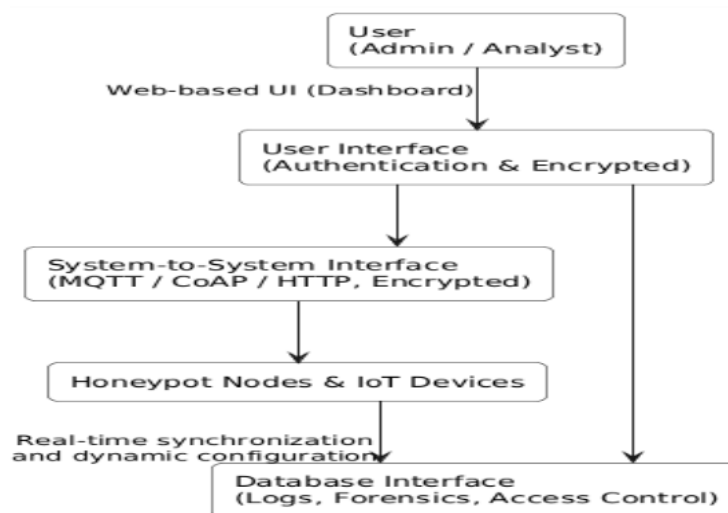


Figure 7 System Interface of the IoT Honeypot & Deception Framework (IHDF) adapted from [35]

Database Interface: The system's logging of attacks, incidents, and the forensic data retrieval process is what the Database Interface does. Data generation comprises a capstone of security intellect report creation. Access to the database is controlled by means of authentication mechanisms in order to keep unauthorized users out and to prevent any possible data exposure. They are not just facilitating the work of IHDF, but also taking care of the whole process from its collection to its visual display, thus, creating a reliable and scalable environment for deception.

4.6 Summary

In Chapter 4, a systematic and thorough analysis was given for the IoT Honeypot & Deception Framework (IHDF) that is being proposed. It started with the difference between the system and the conventional defensive means, such as IDS and firewalls, asserting its proactive and intelligence-led character. The chapter then elaborated the requirements elicitation techniques, consisting of literature review and comparative analysis, which assisted in forming the core objectives of the system. After that, the use-case diagram along with descriptions portrayed the interaction among the system actors, namely the Administrator, Analyst, and Attacker, in the deception setting. Subsequently, the non-functional and functional requirements were specified in such a way that the system's adaptability, scalability, and friendliness were ensured. In the System Interfaces part, the major communication and operation channels in the IHDF were also pointed out, including user interaction, system communication, and secure data storage. In conclusion, the present chapter gave a comprehensive examination for the honeypot framework development and its administration through the specification of the system's roles and actions and the communication between users and systems. The insight acquired in this manner served as a foundation for the next chapter on system design and implementation

Chapter 5

Proposed Solution

5 Proposed Solution

The IoT Honeypot and Deception Framework have the intention of enhancing the detection rate of cyber-attacks on IoT and critical infrastructure through the application of honeypots and deception tools.

In the first phase, the low-interaction honeypots are used to identify general types of attacks like scans and brute-force, and high-interaction honeypots are designed in the form of IoT entities and industrial servers. The IOT Active-Ready environment in the second phase has the use of deception assets like mock admin pages, Honeyfiles containing dummy information, and legitimate-looking credentials, triggering the system when accessed. In the third phase, all the activities performed by the attackers on the deception asset are logged and sent to the backend module, where the actions are saved.

These results feed the threat intelligence and analysis process, where the primary focus includes threats like the use of MQTT unauthorized access, Telnet brute-force, and API abuse, with the goal of enhancing the security of IoT. The purpose of the honeypots is to capture the attacking traffic and notify the IDS, and the IDS, in turn, furnishes the threat intelligence to optimize the honeypots.

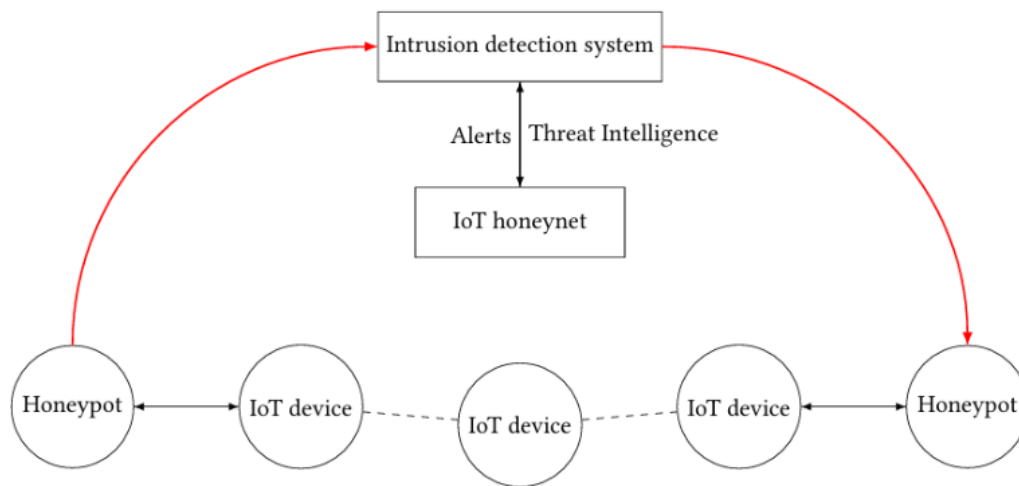


Figure 8 Integration of honeypots with IDS in an IoT network adapted from[15]

The tool enables proactive and intelligence-led responses rather than simply gathering data. Since the typical users cannot access the deception assets, the access signals mean there is malicious activity. There is also a real-time dashboard where the security administrators can

watch the attackers. In other words, the proposed solution enhances the security of IoT systems using the concept of honeypots and deception.

5.1 System Architecture

Network & Communication Architecture

The Network & Communication Architecture forms the first layer of interaction of the IoT Honeypot & Deception Framework with external adversaries. It receives, filters, and routes every incoming network into the protocol emulation components for processing. The architecture begins with the Attacker Node that starts probing the network, login attempts, or malicious payload delivery toward exposed services of the honeypot. All received packets initially pass through the Network Layer, which handles port exposure, traffic forwarding, and basic packet reception while ensuring that only allowed and monitored ports are visible to the attacker.

Once the communication hits the IHDF boundary, it gets processed via a protocol emulation layer that contains dedicated handlers for popular IoT communication protocols, including MQTT, HTTP, Telnet/SSH, and CoAP. Each handler is designed to emulate plausible protocol behavior, device responses, and known vulnerabilities with the purpose of sustaining the deception credibility and engagement time of the attacker.

Finally, all protocol-level interactions are passed into the Honeypot Entry Layer, which initializes attacker sessions, records session metadata, and then securely redirects the communication to core honeypot logic for deep interaction and intelligence gathering. This layered workflow ensures that communication precisely emulates real IoT targets while capturing high-quality data for threat analysis and adaptive deception.

Honeypot & Deception Architecture

The Honeypot & Deception Architecture is the core functional layer for the IoT deception framework-that is, in which real attacker interaction is emulated, captured, and analyzed in a controlled environment. Workflows begin in the Honeypot Containers that are used to

host virtualized, isolated environments that are designed to expose services that are believable in their own right, yet are intended to be vulnerable. These containers dynamically create fake assets and system responses needed to convincingly simulate legitimate IoT device behavior and form the very foundation of a deception strategy. Once these assets have been generated, they are surfaced through the Decoy Services layer, exposing common IoT-facing protocols such as SSH, web interfaces, FTP endpoints, and simulators for industrial services that will attract attackers and maintain realistic engagement.

As attackers begin to interact with the exposed decoys, all activity flows into the Interaction Layer where session-level data is captured at high detail. Events include command execution, login attempts, behavioral sequences, and also attacker decision patterns. This layer acts to maintain high-fidelity behavioral traces while ensuring malicious actions remain contained within the isolated honeypot environment. Suspicious payloads, exploit files, or downloaded binaries in this layer are automatically forwarded to Sandbox for Malware, a controlled execution environment designed and built to analyze unknown or hostile artifacts without threatening the integrity of the system. In this sandbox, the payloads are executed in safe conditions; hence, the framework could observe system level behavior, extract IoCs, and map attacker TTPs.

This is where, in the Threat Intelligence layer, the final stage of processing occurs: the system correlates behavioral logs, sandbox outputs, and session metadata to create actionable insights into the objectives and methods of attackers. The extracted intelligence flows into two main components: the TI Database, with its structured threat record storage capable of long-term analysis and model training; the Alert System triggers when high-risk or novel attack behaviors are detected. Honeypot & Deception Architecture, through this layered approach, provides deep, safe interaction with the attacker to perform complete intelligence gathering while being totally isolated from any production systems.

Backend Analytics & Storage Architecture

Backend Analytics & Storage Architecture is the intelligence core layer of the IoT Honeypot & Deception Framework that converts all captured interactions, session traces, and protocol-level events into valuable security insights. During the communication of the attacker with the honeypot through the Honeypot Entry Layer, the backend processing circuit receives every single event, which performs the actions of log normalization, metadata enrichment, and correlation of attacker behaviors across different sessions. This course of action changes network activity into structured intelligence that is suitable for long-term analysis and adaptive deception strategies. The analytical engine is the most significant component of the whole architecture. It continuously evaluates both the behavior and the threat level of the attacker in real-time, and it also extracts attack signatures. The engine detects not only the presence of the attacker but also the nature of the attack, allowing the honeypot to learn from the attacker, which increases deception believability and thus the engagement time. After all the data has been processed, it is moved to Storage Layer, which is made of time-series databases, session archives, and forensic repositories. This layer keeps all the historical records of the attacks, protocol logs of the device used, and intelligence artifacts, which supports the framework in the areas of deep investigation, long-term trend analysis, and evidence preservation. By means of this systematic backend workflow, the IHDF obtains a dependable and flexible basis for monitoring, analyzing, and comprehending adversarial behavior in IoT environments.

System Architecture Diagram

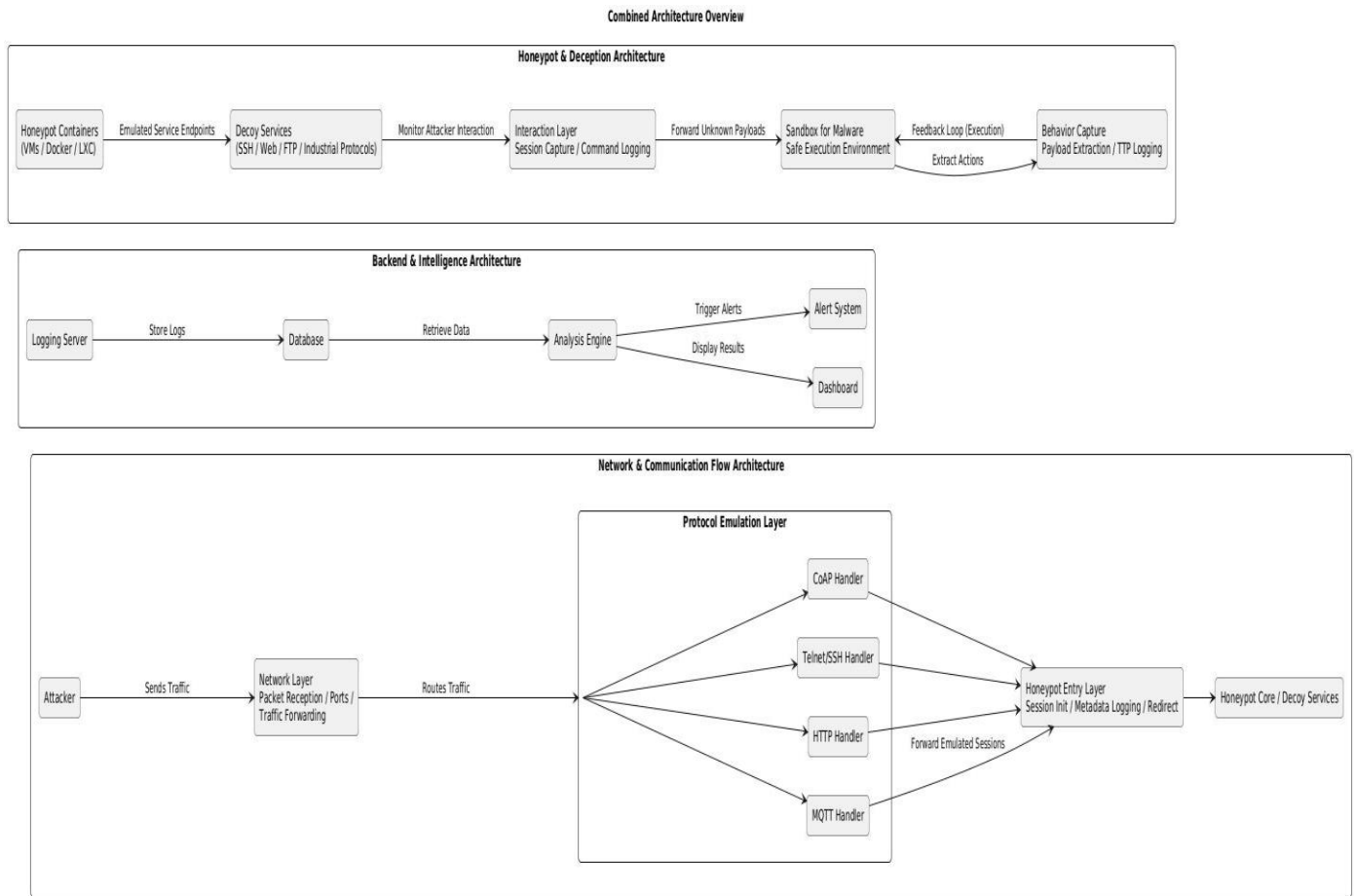


Figure 99 Overall System Architecture of the Proposed IoT Honeypot Framework, showing the integrated deception, analysis, and communication layers.

5.2 ER Diagram

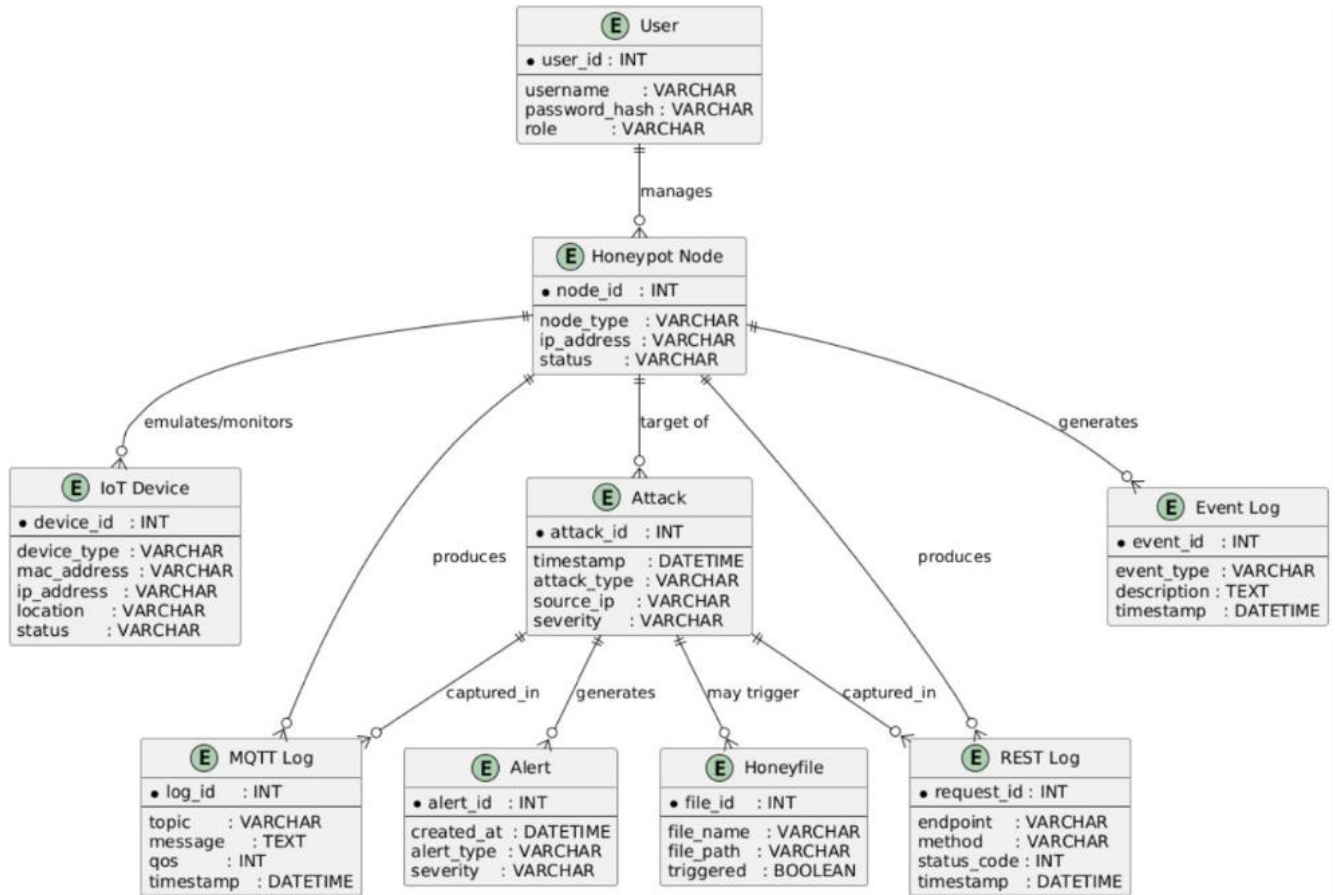


Figure 1010 Entity–Relationship Diagram (ERD) for the IoT Honeypot & Deception Framework

5.3 User Interface Design

This interface represents the main dashboard of the proposed IoT Deception Framework. It provides a high-level overview of the system's security status, including the number of detected attacks, the active honeypots, and the distribution of attack attempts across different IoT protocols. Visual charts are also embedded in the dashboard, enabling the security administrator to visually observe activity trends and analyze attack patterns over time. The design will provide an easy-to-use interface that allows for clear, timely monitoring and response to security events in real-time.

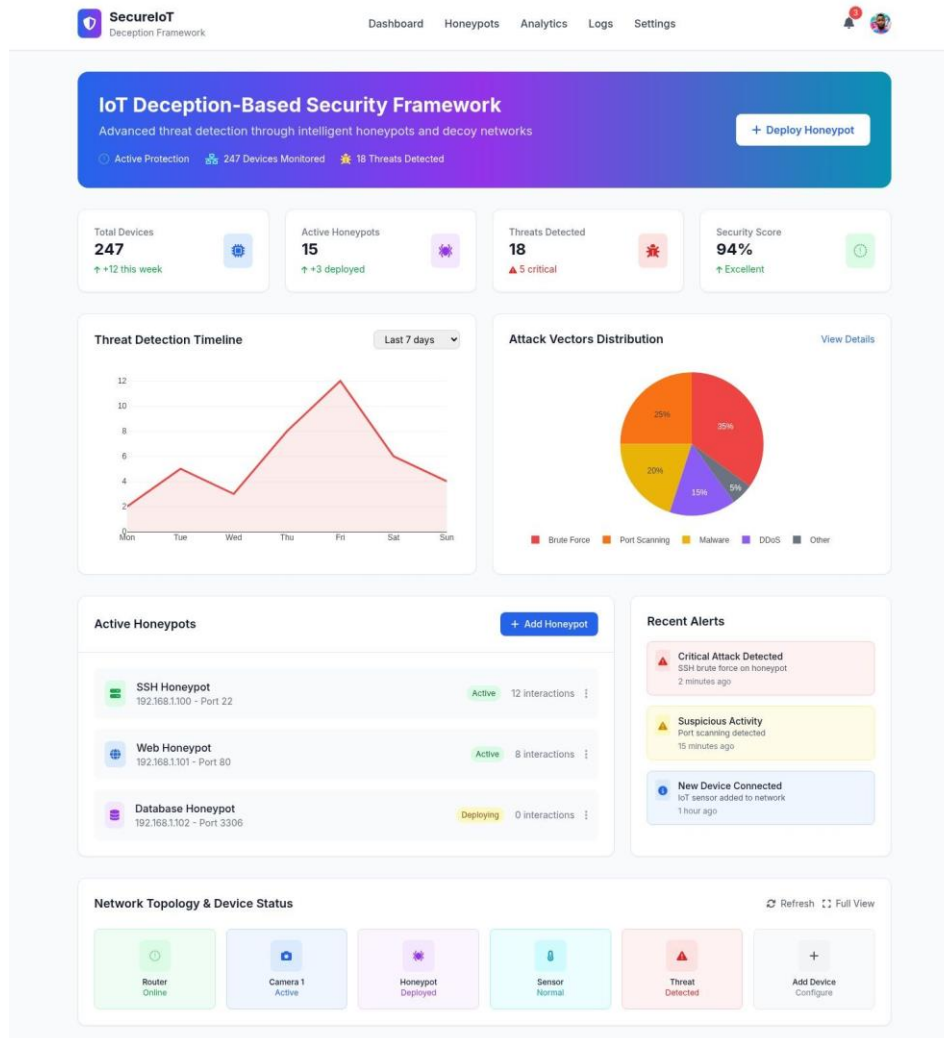


Figure 11 IoT Deception Framework-Home Dashboard

5.4 Summary

The full design of the proposed IoT Honey pot & Deception Framework was presented, describing the solution's layered architecture and core functional components. The chapter first introduced the main idea of the proposed framework, aiming at detecting and understanding the behavior of cyber attackers in IoT environments through a mix of low-interaction and high-interaction honeypots, deception assets, and backend intelligence.

The Network & Communication Architecture described how attacker traffic is received, filtered, and routed into protocol emulation modules that simulate IoT services such as MQTT, HTTP, Telnet/SSH, and CoAP. Honey pot & Deception Architecture detailed how these modules

generate realistic device behaviors while managing attacker sessions and keeping deception credibility. The Backend Analytics & Storage Architecture explained the flow of captured data within the backend processing pipeline, where events go through normalization, enrichment, and correlation before being stored for long-term analysis and the generation of alerts.

The System Architecture Diagram on how all the components interact, and the Entity–Relationship Diagram that defines database structure to store events, logs, alerts, and attacker interactions, were also included in this chapter. Finally, the User Interface Design section has shown how the dashboard visualizes system outputs, allowing analysts to monitor live events, configure decoys, investigate alerts, and export telemetry.

Overall, this chapter presented a holistic view of the IHDF system by explaining how its communication, deception, analytics, and visualization layers cooperate to provide a dependable and intelligent IoT security framework.

Chapter 6

Future Work

6 Future Work

Future work on the IHDF should concentrate on scalability, adaptability, protocol diversity, and real-world evaluation to overcome the limitations identified by previous studies. Various strategic directions are necessary to develop next-generation deception systems in IoT environments, as derived from the literature surveyed.

- **Build Lightweight, Resource-Efficient Honeypots**

Most adaptive and ML-based honeypots have high computational overhead, making them unfit for resource-constrained IoT devices. Future work will consider small-footprint, energy-efficient ML model design capable of running on ARM, MIPS, and RISC-V embedded architectures. Model pruning, quantization, and edge-optimized inference engines will be explored in order to cut down power consumption without sacrificing detection accuracy. This direction aligns with recommendations from studies highlighting the need for lean AI models compatible with IoT hardware.

- **Incorporate Self-Adaptive, Continuous Learning Mechanisms**

State-of-the-art honeypots currently depend highly on static configurations or offline training. Arguably, one of the key improvements in the future could include embedding reinforcement learning, adversarial machine learning, and transfer learning into the technology, thus enabling dynamic adaptation at runtime. The framework will evolve the deception behavior based on attacker interactions--similar to HoneyIoT's MDP-based approach--and continuously retrain from live telemetry. This will reduce the detectability of honeypots while improving their resilience against rapidly changing threats, including botnet propagation and APT campaigns.

- **Expand Protocol and Device Coverage**

The state-of-the-art deception frameworks primarily focus on common protocols like HTTP, MQTT, CoAP, and Telnet. In order to enhance the realism and include a wide range of attack vectors, future work will also extend the support to Zigbee, BLE, proprietary IoT protocols, industrial ICS/SCADA interfaces, and multi-device interactions. This enables the IHDF to more accurately emulate heterogeneous environments, complementing gaps related to limited protocol diversity.

- **Develop Cooperative, Distributed Honeynets**

Various studies demonstrate the efficiency of multi-node, container-based honeypots in capturing lateral movement and coordinated attacks. Future research will involve studying distributed honeynets using Docker/Kubernetes, where several virtual IoT devices can communicate and react in synchronized patterns to simulate realistic IoT ecosystems and enable better detection of multi-stage intrusions. Scalability and remote manageability can be further enhanced with cloud-based deployment models.

- **Standardized Evaluation and Metrics Frameworks**

The literature highlights a significant gap in consistent methods of evaluation. Thus, future work will involve the design of standardized performance metrics with respect to deception effectiveness, attacker engagement time, detectability resistance, and model efficiency. Such metrics will allow comparing multiple deception configurations and validating improvements scientifically across iterations.

- **Perform Large-Scale, Long-Term Real-World Experiments**

Most prior works rely on laboratory simulations, leaving open questions about real deployment performance. The IHDF project will perform extensive field experiments in realistic IoT networks (smart homes, small industrial setups, and campus IoT). Such deployments will measure long-term scalability, stability, resilience, and the ability to handle sustained attack traffic under real conditions.

- **Improve data analytics, storage efficiency, and automation**

Because IoT honeypots produce very large datasets, future work will involve the integration of automated log correlation, ML-based anomaly clustering, and threat intelligence extraction pipelines. These systems will transform honeypot data into actionable detection rules for IDS/IPS systems. Ultimately, efficient storage and distributed data analytics will be introduced to handle high-volume telemetry without taxing limited IoT resources.

- **Enhancing Realism and Reducing Honeypot Detectability**

- Attackers are increasingly capable of identifying honeypots. Future enhancements will include:
- randomized system fingerprints,

- dynamic service mutation,
- realistic timing behavior,
- Emulation at the firmware level using tools such as Honware.
- This will enhance credibility and engagement to ensure that attackers cannot easily bypass the system.

Interoperability Support among IoT Platforms

One of the repeated limitations of previous works is the lack of compatibility with various platforms. Future work should shift to the design of interoperable deception modules that will be seamlessly integrated with cloud IoT providers, fog/edge nodes, and hybrid architectures. The flexibility will make IHDF more applicable in various deployment scenarios.

6.1 Project Timeline

Below is the suggested timeline for project second phase (Graduation Project 2). It provides a general timeline of major development tasks as well as a particular assignment of responsibilities for each team member. It is planned step by step to enable continuous progress and complete all

milestones efficiently within the semester.

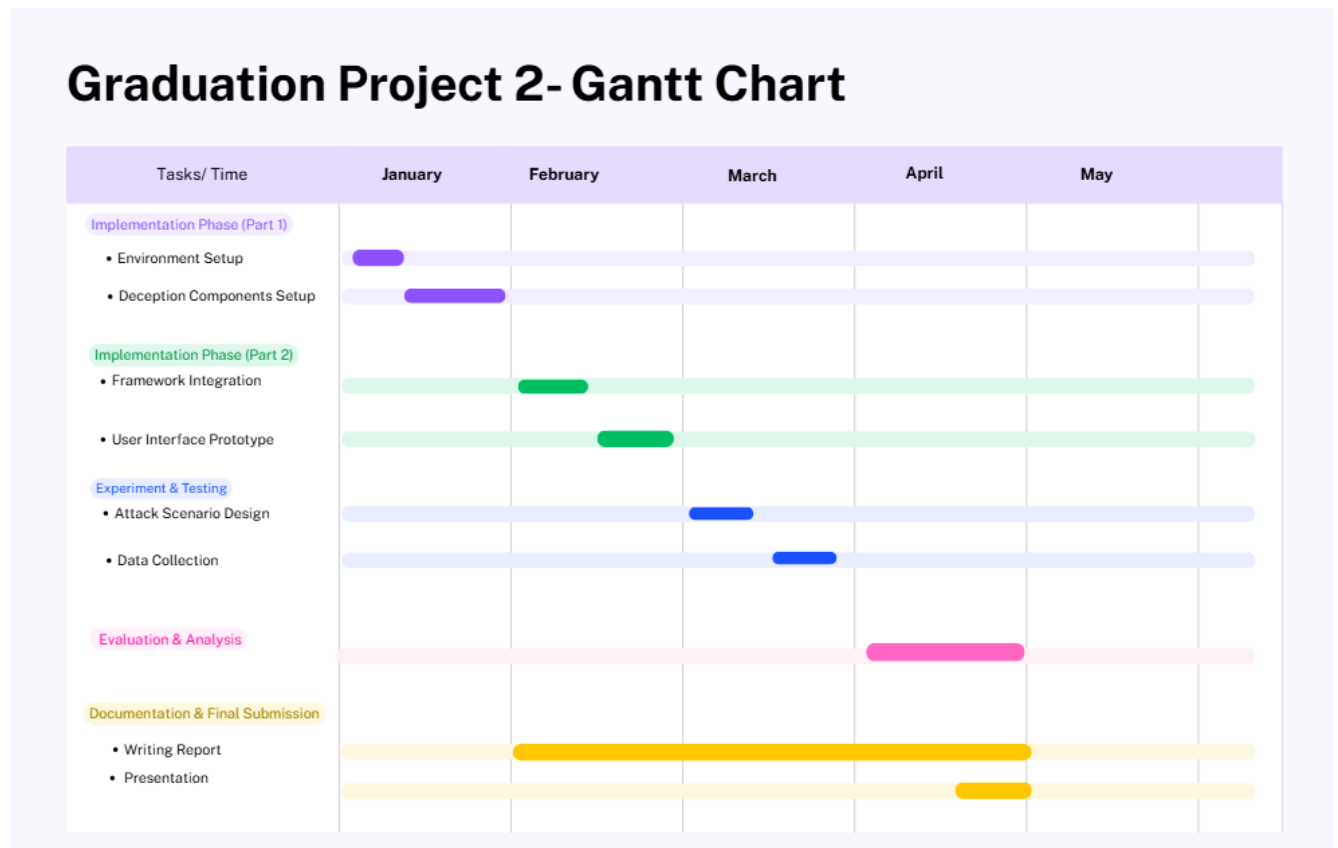


Figure 12 12Project Timeline (Gantt Chart)

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